# IAT New Test 18

Question paper, answer key and worked solutions

**60 questions 180 minutes 240 marks**

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

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**Instructions**

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

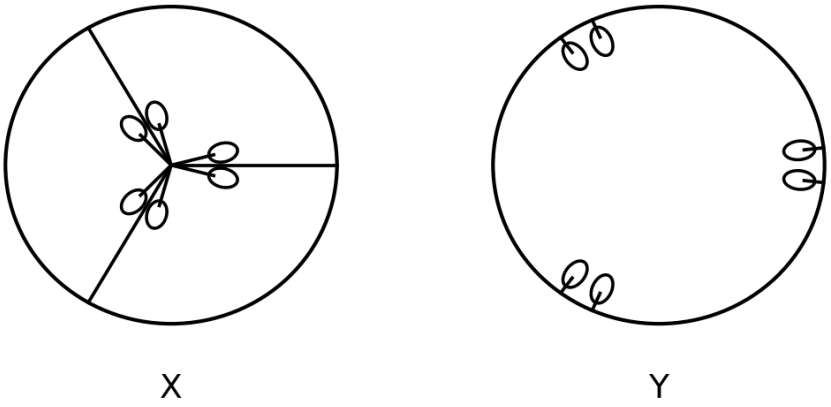
# Biology

**Q1** In a basidiomycete, two compatible haploid hyphae fuse their cytoplasm, but the paired nuclei remain separate through many divisions. Later, two nuclei fuse in a basidium and meiosis follows. Which sequence correctly describes the nuclear states and the source of variation?
**A)** Diploid hyphae persist; mitosis in the basidium halves ploidy
**B)** Dikaryotic hyphae persist; a short diploid stage precedes meiotic haploid products
**C)** Haploid hyphae persist with one nucleus; plasmogamy itself creates diploid nuclei
**D)** Dikaryotic hyphae are tetraploid; karyogamy restores diploidy

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**Q2** The figure shows transverse sections of two ovaries. In X, septa meet a central column bearing ovules; in Y, ovules attach to the inner outer-wall surface and septa are absent. Which pair of placentation types fits X and Y, respectively?

**Ovary cross-sections: X and Y**




Lines: septa; small ovals: ovules

Outer circle: ovary wall

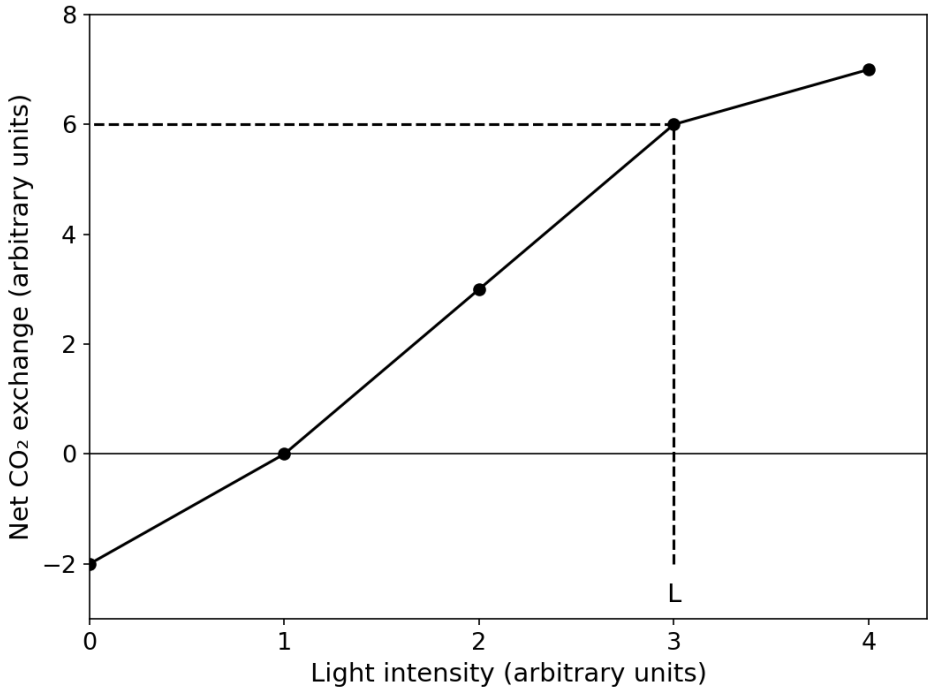
**A)** Axile and parietal
**B)** Parietal and axile
**C)** Free central and marginal
**D)** Basal and free central

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**Q3** A purified double-stranded DNA sample is treated with an enzyme that hydrolyses phosphodiester bonds but leaves hydrogen bonds initially undisturbed. Which immediate chemical change is expected?
**A)** Purines become pyrimidines
**B)** Peptide bonds in the bases break
**C)** The sugar-phosphate backbone is cut without directly hydrolysing base-pair hydrogen bonds
**D)** All deoxyribose becomes ribose

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**Q4** The figure plots net CO2 exchange of one leaf against light intensity. Positive values mean net uptake. Assume respiratory CO2 release stays constant across the plotted range and the dark value is -2 units. At light level L, net uptake is 6 units. What are gross fixation at L and net carbohydrate gain at the compensation point?




**A)** 6 units and 2 units
**B)** 8 units and zero
**C)** 4 units and zero
**D)** 8 units and 2 units

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**Q5** A kidney preparation retains normal ascending-limb NaCl transport, but blood flow through the vasa recta is made so rapid that the medullary osmotic gradient is largely washed out. ADH is high and the collecting duct remains water-permeable. Which outcome is expected relative to an intact gradient?
**A)** More concentrated urine because high ADH supplies the osmotic force
**B)** Unchanged concentration because ascending-limb transport is normal
**C)** No glomerular filtration because medullary blood moves faster
**D)** Less concentrated urine because water permeability cannot replace the interstitial driving gradient

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**Q6** A normal diploid angiosperm has 2n=18. An unreduced embryo sac forms without meiosis and contains a 2n egg and two 2n polar nuclei. The egg develops parthenogenetically, but the central cell is fertilised by one normal haploid sperm. What chromosome numbers occur in the embryo and primary endosperm nucleus?
**A)** 18 and 45
**B)** 27 and 27
**C)** 18 and 27
**D)** 36 and 45

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**Q7** In a large population, genotype frequencies at one autosomal locus are AA=0.50, Aa=0.20 and aa=0.30. For one generation, gametes unite randomly with no selection, migration or mutation and equal fertility of genotypes. What is the expected change in heterozygote frequency?
**A)** An increase from 0.20 to 0.24
**B)** A decrease from 0.20 to 0.16
**C)** An increase from 0.20 to 0.48
**D)** No change because allele frequencies are conserved

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**Q8** A bacterial population grows on nutrient medium. Antibiotic added to a matched culture greatly reduces viable count, but a few colonies survive. Removing antibiotic lets both sensitive and resistant isolates grow. Which conclusion is supported without assuming the antibiotic is a nutrient?
**A)** Every sensitive bacterium must have converted to resistance
**B)** Resistance can provide a conditional advantage without being necessary for growth in drug-free medium
**C)** Survivors cannot reproduce without antibiotic
**D)** Resistance proves a larger growth rate under every environmental condition

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**Q9** A PCR assay has a target-specific primer pair. The known-positive tube produces the expected band, the sample tube produces that band, and a no-template tube containing the same reagent mix also produces that band. Which interpretation is justified before repeating with clean reagents?
**A)** The sample definitely lacks the target
**B)** The polymerase is inactive
**C)** The band in the no-template tube proves spontaneous assembly of the target sequence
**D)** Contamination prevents assigning the sample band uniquely to sample DNA

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**Q10** Over the same interval, two lake communities have equal gross primary production. In lake X, producer respiration is higher, while measured herbivore secondary production is also higher. Which conclusion necessarily follows, without assuming equal transfer efficiency?
**A)** X has lower net primary production, but the higher secondary production does not contradict that
**B)** X must have higher net primary production because herbivores grow faster
**C)** The measurements are impossible by energy conservation
**D)** Producer respiration must equal herbivore production in each lake

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**Q11** Large replicate populations initially contain the same heritable colour variants. In habitat H, dark individuals survive better than light individuals; in habitat J the survival ranking reverses. Mating and fertility are otherwise equal. Which observation would most directly undermine survival selection as the explanation for a rise of dark colour in H?
**A)** Dark colour is inherited by offspring
**B)** Dark colour was present before the habitat treatment
**C)** Dark frequency rises equally in a matched no-mortality H control with the same mating and no migration or mutation
**D)** Light individuals survive better in J

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**Q12** A recessive lethal allele aa dies before birth, whereas AA and Aa survive equally. At an independent locus, B is completely dominant to b and all genotypes survive. Parents AaBb and AaBb mate. Among live-born offspring, what fraction is heterozygous at both loci?
**A)** 1/4
**B)** 3/8
**C)** 1/2
**D)** 1/3

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**Q13** In a stated ideal axon model, an initiated action potential travels in both directions at 20 m/s through excitable membrane. Each passed point is absolutely refractory for 2 ms. Electrodes P and Q are 2 cm apart on an otherwise uniform axon. P initiates a pulse at t=0; Q is stimulated at t=0.5 ms. The stimuli act only at their sites. What happens to the two inward-moving wavefronts?
**A)** They collide at 1.5 cm from P at 0.75 ms and do not pass through each other
**B)** They pass through each other and reach the opposite electrodes unchanged
**C)** Q cannot initiate because the first wave reached Q before 0.5 ms
**D)** They collide halfway at 0.5 ms and merge into a faster wave

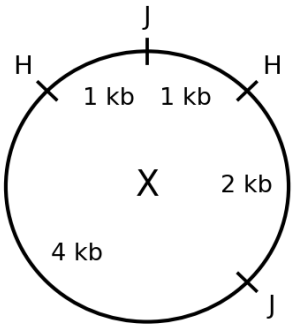
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**Q14** Red blood cells are placed in two solutions with the same initial osmolarity as their cytoplasm. Solute X cannot cross the membrane; solute Y crosses rapidly, and water crosses rapidly in both trials. Initially intracellular X and Y are negligible. Ignore metabolism and ion transport. Which comparison is expected after equilibration begins?
**A)** Both solutions permanently preserve cell volume because initial osmolarities match
**B)** X preserves volume, whereas Y entry produces swelling and may cause lysis
**C)** Y preserves volume, whereas X must enter by active transport
**D)** Both solutions force immediate water loss because solute concentration is nonzero

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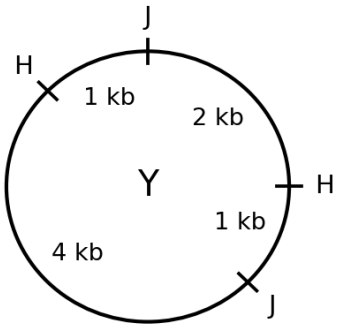
**Q15** A circular 8 kb recombinant vector has unique sites E and H 1 kb apart in its empty 5 kb backbone. A 3 kb insert is placed at E; its unique internal H site is 1 kb from one end and 2 kb from the other. The figure shows both possible insert orientations and the unchanged backbone H. A complete H-only digest is run. Which fragment pair distinguishes orientation X, whose insert H lies 1 kb along the E-to-insert path shown?

**Circular recombinant maps: junctions came from E**




Insert J–J: 3 kb

Backbone: 5 kb




Insert J–J: 3 kb

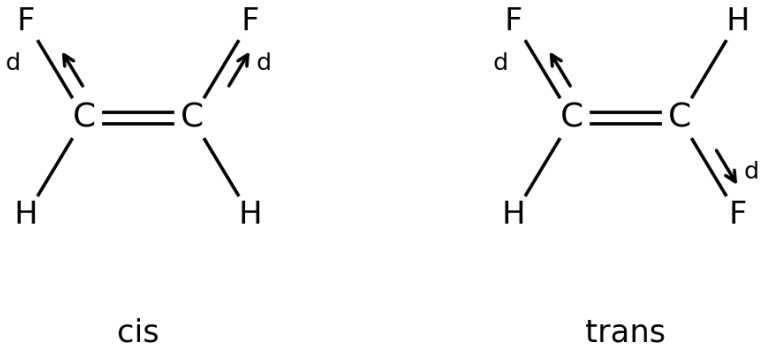
Backbone: 5 kb

**A)** 1 kb and 7 kb
**B)** 2 kb and 3 kb
**C)** 2 kb and 6 kb
**D)** 3 kb and 5 kb

# Chemistry

**Q16** The figure shows cis- and trans-1,2-difluoroethene with ideal trigonal-planar carbon geometry. In a stated bond-vector model, each C-F bond contributes a dipole of magnitude d towards F; neglect all other bond contributions. Which net dipole magnitudes follow for cis and trans, respectively?

**Ideal planar C–F bond-vector model**




**A)** $d, 0$
**B)** $2d, d$
**C)** $\sqrt{3}d, 0$
**D)** $0, \sqrt{3}d$

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**Q17** Two Ni(II) complexes have four ligands each. X is square-planar $[\mathrm{Ni(CN)_4}]^{2-}$; Y is tetrahedral $[\mathrm{NiCl_4}]^{2-}$. Nickel has Z=28. Under the usual valence-bond description, which magnetic comparison is correct?
**A)** X has two unpaired electrons; Y none
**B)** Both have no unpaired electrons because coordination number is four
**C)** Both have four unpaired electrons because Ni(II) is d6
**D)** X has no unpaired electrons; Y has two

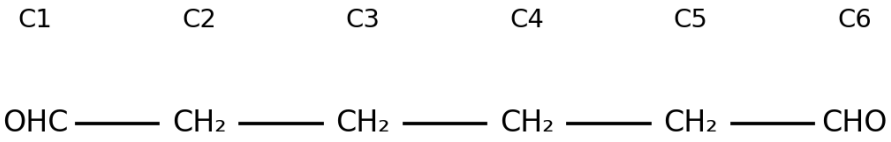
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**Q18** For a reaction, assume $\Delta H^\circ = +20$ kJ/mol and $\Delta S^\circ = +50$ J/(mol K) are constant. At 300 K take RT=2.5 kJ/mol. Compare the forward reaction at dimensionless quotients $Q_1 = e^{-4}$ and $Q_2 = e^{-1}$. Which statement is correct?
**A)** Forward is favourable at Q1 but unfavourable at Q2
**B)** Forward is unfavourable at both because the standard Gibbs change is positive
**C)** Both mixtures are at equilibrium because temperature is fixed
**D)** Forward is favourable at Q2 but unfavourable at Q1

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**Q19** The figure shows hexane-1,6-dial, $OHC-(CH_2)_4-CHO$, numbered C1 through C6. Under controlled intramolecular aldol addition, the enolate at C2 attacks carbonyl C6; ignore competing reactions and stop before dehydration. What ring size and retained carbonyl group result?

**Hexane-1,6-dial: labelled carbon positions**




**A)** Six-membered ring with no carbonyl
**B)** Five-membered ring with an aldehyde substituent
**C)** Four-membered ring with a ketone substituent
**D)** Five-membered ring with two unchanged aldehydes

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**Q20** A copper sulfate solution is electrolysed with pure copper electrodes. Assume only $Cu^{2+} + 2\,e^- \rightleftharpoons Cu$ operates at each electrode, equal current efficiency, constant volume and no side reactions. Which changes occur?
**A)** Both electrodes gain mass and Cu2+ falls
**B)** The cathode dissolves and Cu2+ rises
**C)** The anode loses the mass gained by the cathode and bulk Cu2+ stays constant
**D)** Both electrodes lose mass with no ion transfer

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**Q21** Benzamide is treated with bromine and aqueous NaOH under Hofmann bromamide conditions. The isolated organic product is then tested with chloroform and alcoholic KOH. Which result is expected?
**A)** Benzylamine forms and loses its nitrogen in the second test
**B)** Aniline forms and gives the carbylamine reaction
**C)** Benzoic acid forms and gives the carbylamine reaction
**D)** N-methylaniline forms and gives the carbylamine reaction

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**Q22** An acidic sample containing 0.120 mol Fe2+ reacts completely with 0.030 mol dichromate. Each dichromate accepts six electrons to form Cr3+. Residual dichromate is then reduced by excess iodide to iodine, and the iodine is titrated by $I_2 + 2\,S_2O_3^{2-} \longrightarrow 2\,I^- + S_4O_6^{2-}$. How many moles of thiosulfate are required? Assume no side reactions.
**A)** 0.020
**B)** 0.030
**C)** 0.180
**D)** 0.060

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**Q23** A reversible elementary reaction has forward activation energy 40 kJ/mol and reaction enthalpy -20 kJ/mol. Assume both directions obey Arrhenius behaviour with temperature-independent activation energies and pre-exponential factors. Between T1 and T2, $\ln[k_f(T_2)/k_f(T_1)] = 2$. What is $\ln[k_r(T_2)/k_r(T_1)]$?
**A)** 3
**B)** 1
**C)** 2
**D)** 4

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**Q24** Methyl benzoate reacts with excess ammonia to give an amide. That isolated amide undergoes complete acidic hydrolysis. What are the principal organic products after the first and second stages, respectively?
**A)** Aniline and benzene
**B)** Benzylamine and benzyl alcohol
**C)** Benzamide and benzoic acid
**D)** Acetamide and phenol

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**Q25** Compare drawings P: $CH_3CHO$ and Q: $CH_2{=}CHOH$. They have the same molecular formula but differ in a hydrogen's attachment and bond positions. Which description is correct?
**A)** They are resonance contributors because all atoms remain in identical positions
**B)** They are different molecular formulae because a double bond moves
**C)** They are geometrical isomers about the same double bond
**D)** They are tautomers; interconversion requires proton and bonding changes

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**Q26** A solution containing a fixed amount of fully dissociated nonvolatile salt initially has osmotic pressure pi0 and volume V0. An ideal reverse-osmosis membrane passes only water into a pure-water reservoir. A constant external pressure difference 3pi0 compresses the solution side. Assume constant temperature, ideal behaviour, no salt leakage and volume proportional to remaining water. What fraction of the original water can be removed at the limiting equilibrium?
**A)** 1/3
**B)** 2/3
**C)** 3/4
**D)** 1/2

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**Q27** A hypothetical element has only two isotopes with masses 63 u and 65 u. Its mean atomic mass is 63.6 u. A representative sample contains 0.0500 mol of atoms. How many moles of the heavier isotope are present?
**A)** 0.0150 mol
**B)** 0.0350 mol
**C)** 0.0300 mol
**D)** 0.0250 mol

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**Q28** For the process $X_2(g) \longrightarrow 2\,X^-(aq)$, consider the enthalpy per mole of X atoms. Atomisation costs are +80 kJ for F and +120 kJ for Cl; electron-gain enthalpies are -330 and -350 kJ; hydration enthalpies are -500 and -360 kJ, respectively. Which conclusion follows from this cycle alone?
**A)** Cl is favoured by 160 kJ because its electron gain is more exothermic
**B)** F is favoured by 20 kJ because only electron gain matters
**C)** Both have equal overall changes because both gain one electron
**D)** F is favoured by 160 kJ despite Cl's more exothermic isolated electron gain

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**Q29** An amino acid has two carboxyl groups with pKa values 2 and 4, and one ammonium group with pKa 9. Treat each group as predominantly protonated below and deprotonated above its pKa. At pH 6, what predominant net charge and electrophoretic migration are expected?
**A)** Zero charge; no directional migration
**B)** +1; towards the cathode
**C)** -1; towards the anode
**D)** -2; towards the cathode

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**Q30** Three identical aliquots each contain an unknown amount n of monoprotic weak acid HA. After adding 5.00 mmol NaOH to one aliquot and diluting to 200 mL, pH is $4 - \log_{10} 3$. After adding 15.00 mmol to another and diluting to 200 mL, pH is $4 + \log_{10} 3$. Both are buffers. A third aliquot is exactly neutralised and diluted to 200 mL. At 25 C, estimate its pH using Kw=10^-14 and the usual small-ionisation approximations.
**A)** 8.50
**B)** 4.00
**C)** 7.00
**D)** 9.00

# Mathematics

**Q31** A circle passes through A=(0,1) and B=(2,3), and its centre lies on y=x. What is the length of a tangent segment drawn from the origin to the circle?
**A)** $\sqrt{2}$
**B)** 2
**C)** $\sqrt{5/2}$
**D)** $3/\sqrt{2}$

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**Q32** An axis-aligned ellipse centred at the origin has equation $x^2/a^2 + y^2/b^2 = 1$, with a,b>0. A chord has midpoint (1,1), slope -2 and length $2\sqrt{5}$. What is $a^2 + b^2$?
**A)** 9
**B)** 9/2
**C)** 18
**D)** 27/2

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**Q33** What is the total area between $y = 1$ and $y = |x^2 - 1|$ over the intervals where $|x^2 - 1| \le 1$?
**A)** $8\sqrt{2}/3$
**B)** $8(\sqrt{2} - 1)/3$
**C)** $4(\sqrt{2} - 1)/3$
**D)** $2/3$

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**Q34** Evaluate $\int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin^4 x}\, dx$.
**A)** $\pi/4$
**B)** $1/2$
**C)** $\pi/8$
**D)** $1$

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**Q35** Let $f(x) = x + \ln x$ for x>0, and let g be its inverse. What is $g''(1)$?
**A)** 1/8
**B)** -1/8
**C)** 1/4
**D)** -1/4

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**Q36** For real a,b with a nonzero, let $f(x) = (ax + b)/(x - a)$. Suppose f(0)=2 and f(1)=3. On its natural real domain, which statement is correct?
**A)** f has one real fixed point and no inverse
**B)** f is its own inverse and has no real fixed point
**C)** f is its own inverse and has two real fixed points
**D)** f is constant because its numerator and denominator are linear

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**Q37** Let $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ and $B_t = I + A + tA^2$, with real t. Over all t for which Bt is invertible and $\mathrm{tr}(B_t^{-1}) = -3/4$, what is the sum of those t values?
**A)** 1
**B)** -2
**C)** 2
**D)** -1

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**Q38** Three distinct tickets are selected uniformly from labels 1 through 7. Given that the largest minus the smallest selected label is 4, what is the probability that label 4 is selected?
**A)** 1/3
**B)** 3/7
**C)** 1/9
**D)** 2/3

**Q39** What is the sum of all x in $(0, \pi)$ satisfying $\tan x + \cot x = 4$?
**A)** $\pi$
**B)** $\pi/4$
**C)** $\pi/2$
**D)** $3\pi/2$

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**Q40** A plane contains the intersection line of $x + y = 1$ and $y + z = 1$, and is perpendicular to $2x - y + z = 0$. What is the distance from (2,1,4) to that plane?
**A)** $\sqrt{2}$
**B)** $4$
**C)** $3/\sqrt{2}$
**D)** $2\sqrt{2}$

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**Q41** Nonzero vectors a and b satisfy $\text{proj}_{\mathbf{b}}\,\mathbf{a} = \mathbf{b}$ and $\text{proj}_{\mathbf{a}}\,\mathbf{b} = \mathbf{a}/4$. What is $|\mathbf{a} + \mathbf{b}|/|\mathbf{a} - \mathbf{b}|$?
**A)** $\sqrt{3}$
**B)** $\sqrt{7/3}$
**C)** $7/3$
**D)** $1$

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**Q42** A complex number z satisfies $|z| = 1$ and $|z - 1||z + 1| = 1$. What value must $z^6$ have?
**A)** 1
**B)** $i$
**C)** $-i$
**D)** -1

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**Q43** For how many integers k in $1 \le k \le 12$ does $x^2 - kx + k + 3 = 0$ have two distinct positive roots r,s satisfying $|1/r - 1/s| > 1/2$?
**A)** 1
**B)** 2
**C)** 3
**D)** 4

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**Q44** What is the constant term in $(x^2 + x^{-1})^8(x - x^{-1})^6$?
**A)** 0
**B)** 420
**C)** -420
**D)** 840

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**Q45** A real geometric progression has $|r| < 1$. The sum of all its terms is 12, and the sum of its odd-indexed terms is 9. What is the sum of the squares of all its terms?
**A)** 36
**B)** 72
**C)** 81
**D)** 144

# Physics

**Q46** A small bead of mass m moves on the inside of a smooth vertical circular track of radius R. At the top, its speed is exactly $\sqrt{gR}$. What is the normal force there, and what supplies the inward acceleration?

**A)** mg; the normal force alone
**B)** Zero; gravity supplies the inward acceleration
**C)** Zero; the inward acceleration is also zero
**D)** 2mg; gravity and normal force act oppositely

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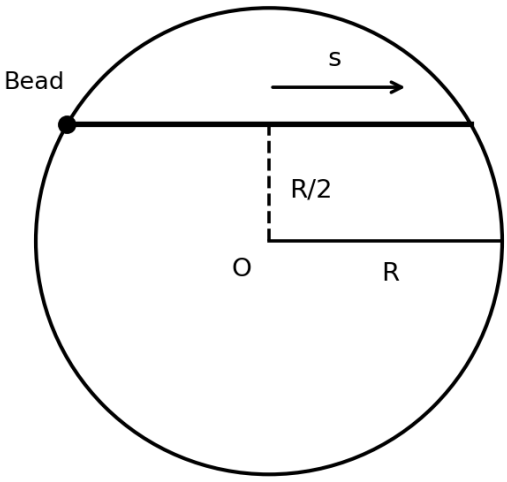
**Q47** A 1.0 kg lump moving at 6.0 m/s embeds in a stationary 2.0 kg block at the instant the block touches an uncompressed horizontal spring of constant 20 N/m. During the brief collision external impulse is negligible. Afterward the joined mass compresses the spring against a constant opposing friction force 4.0 N. What is the maximum compression?

**A)** 1.0 m
**B)** 0.30 m
**C)** 0.60 m
**D)** $\sqrt{0.60}$ m

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**Q48** A narrow straight tunnel is cut through a uniform spherical planet of radius R and surface gravity g. Its closest approach to the centre is R/2 as shown. Neglect removed mass, rotation and friction. A bead is released from rest at one surface entrance. Compared with a diameter tunnel in the same planet, what are its oscillation period ratio and maximum speed ratio?

**Straight tunnel through a uniform spherical planet**




**A)** $1/2, 1/2$
**B)** $\sqrt{3}/2, 1$
**C)** $1, 1/2$
**D)** $1, \sqrt{3}/2$

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**Q49** Two wires of equal length L and Young modulus Y are joined in series. Their cross-sectional areas are A and 2A. Both remain linearly elastic up to the same limiting tensile stress sigma. Neglect weight. What total elastic energy is stored just as the first wire reaches that limiting stress?
**A)** $3\sigma^2 AL/(4Y)$
**B)** $\sigma^2 AL/(2Y)$
**C)** $3\sigma^2 AL/(2Y)$
**D)** $\sigma^2 AL/(4Y)$

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**Q50** Two rods of equal length and area, with conductivities k1 and k2, form a steady series path between 100 C and 0 C. Rod 1 touches the hot end, and their junction is at 40 C. They are then rearranged as parallel paths between the same reservoirs, with ideal contacts and no lateral loss. What is the ratio of the new total heat current to the original current?
**A)** 5/2
**B)** 6/25
**C)** 25/6
**D)** 5/6

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**Q51** An ideal gas with molar Cv=3R is compressed reversibly and adiabatically, with molecule number and collision diameter fixed. Its mean free path falls to one-eighth of its initial value. In the dilute hard-sphere model, what are the final-to-initial pressure ratio and mean time between collisions ratio? Use rms speed as the speed scale for that time.
**A)** $8, 1/8$
**B)** $16, 1/(8\sqrt{2})$
**C)** $2, 1/\sqrt{2}$
**D)** $16, \sqrt{2}/8$

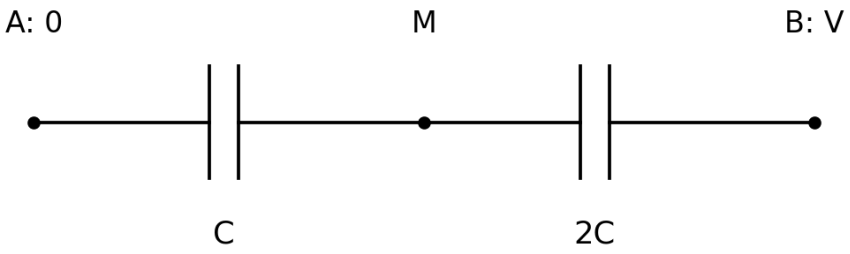
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**Q52** A pendulum of length L oscillates at small angular amplitude theta0. At its lowest point its string is rapidly shortened to L/4 by a force directed along the string. During the shortening gravity's impulse is negligible and angular momentum about the pivot is conserved. Assume the new amplitude is also small. What is the new angular amplitude?
**A)** $2\theta_0$
**B)** $4\theta_0$
**C)** $\theta_0/2$
**D)** $8\theta_0$

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**Q53** The circuit has a capacitor C between grounded node A and isolated junction M, and capacitor 2C between M and node B held at potential V>0. Junction M retains net free charge +CV; it has no conducting path to A or B. Neglect stray capacitance. What are M's potential and the total capacitor energy?

**Two capacitors with an isolated charged junction**




M isolated: $Q_m = +CV$

**A)** $2V/3,\ CV^2/3$
**B)** $V/3,\ CV^2/2$
**C)** $V,\ CV^2/2$
**D)** $V,\ 3CV^2/2$

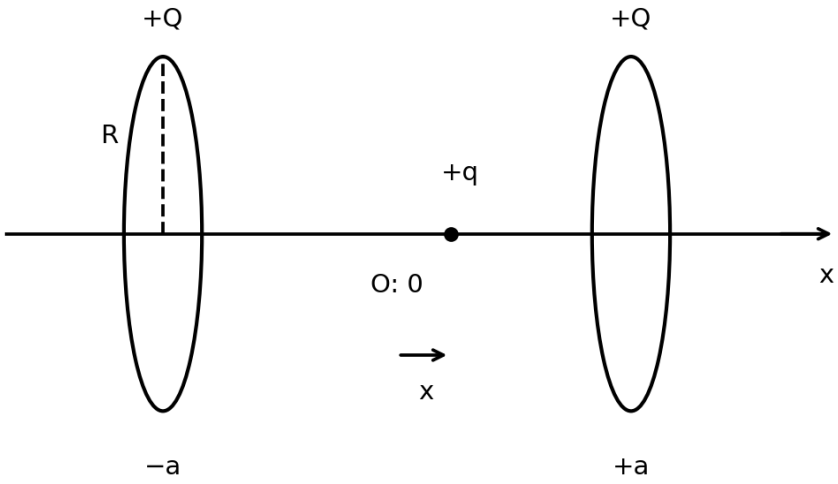
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**Q54** A rigid current loop has fixed magnetic moment magnitude mu and is initially aligned with a uniform field of magnitude B. The field is reversed instantly without giving the loop a mechanical impulse. Immediately afterward, what are its potential energy, torque magnitude and stability against a small angular displacement?
**A)** $+\mu B, 0,$ unstable
**B)** $-\mu B, 0,$ stable
**C)** $+\mu B, \mu B,$ stable
**D)** $0, \mu B,$ unstable

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**Q55** Two identical rings of radius R, each carrying fixed charge +Q, have common x axis and centres at x=+a and x=-a. A positive test charge is constrained to move only on this axis. For one ring, its axial field at displacement s is $kQs/(R^2+s^2)^{3/2}$. For which a is a small displacement from x=0 opposed by a nonzero linear restoring force?

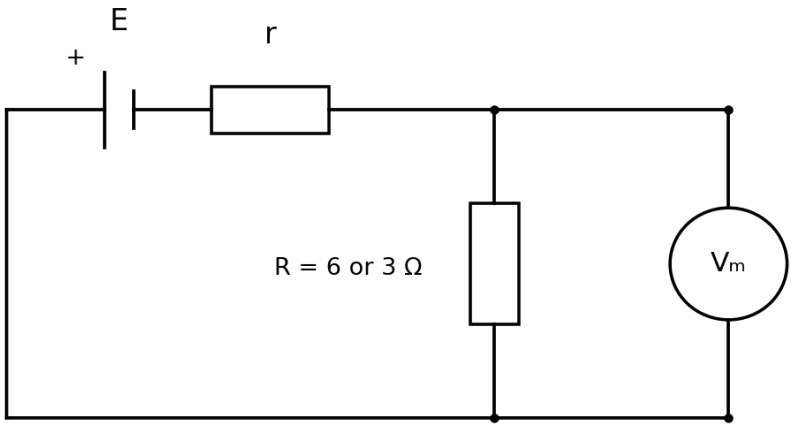
**Equal positive rings on a common axis**




**A)** $a < R/\sqrt{2}$
**B)** $a = 0$ only
**C)** $a = R/\sqrt{2}$ only
**D)** $a > R/\sqrt{2}$

**Q56** A cell of unknown emf E and internal resistance r drives an external load R. A voltmeter of resistance 6 ohm remains across the load. It reads 6 V when R=6 ohm and 5 V when R=3 ohm. The meter is then removed. What maximum power can the cell deliver to a positive variable load?

**Cell with a finite-resistance voltmeter**




Meter: 6 Ω

**A)** 6.25 W
**B)** 12.5 W
**C)** 25 W
**D)** 50 W

---

**Q57** A thin converging lens forms an inverted real image with magnification magnitude 2. The object moves 6 cm towards the stationary lens, after which the image is upright and virtual with magnification 4. What is the lens focal length?
**A)** 4 cm
**B)** 6 cm
**C)** 12 cm
**D)** 8 cm

---

**Q58** At one frequency above both thresholds, photoelectrons from metals A and B have maximum kinetic energies 2.0 eV and 0.5 eV. The frequency is then increased equally for both metals. Which statement is correct while both still emit?
**A)** B’s work function exceeds A’s by 1.5 eV, and the stopping-potential difference stays 1.5 V
**B)** A’s work function exceeds B’s by 1.5 eV, and the difference stays 1.5 V
**C)** The work functions are equal, and the difference must vanish
**D)** B’s work function exceeds A’s by 2.5 eV, and the difference doubles

---

**Q59** An alpha particle of kinetic energy 7.2 MeV approaches a fixed nucleus of charge +50e head-on. Treat the target as infinitely massive and use $ke^2 = 1.44$ MeV fm. Rutherford's Coulomb model remains valid down to closest approach. What distance is reached, and what does it establish about the nuclear radius?
**A)** 10 fm; the radius is exactly 10 fm
**B)** 20 fm; the radius is exactly 20 fm
**C)** 20 fm; it supplies an upper scale bound, not an exact radius measurement
**D)** 40 fm; the radius must exceed 40 fm

---

**Q60** A 2.0 H ideal inductor carries 3.0 A when its source is disconnected. It remains in a closed series loop containing only resistors 2.0 ohm and 6.0 ohm. Assume negligible radiation and no other energy storage. After the current has decayed to zero, how much heat appeared in the 6.0 ohm resistor?
**A)** 9.0 J
**B)** 6.75 J
**C)** 2.25 J
**D)** 18 J

# Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

## Biology

**Q1 / B** Delayed karyogamy versus nuclear counting in a fungal life cycle

**Answer:** (B) Dikaryotic hyphae persist before a brief diploid stage.

**Why:** Plasmogamy joins cytoplasm without immediately joining the two haploid nuclei. A dikaryon therefore has two distinct haploid nuclear lineages, not a single diploid nucleus, and mitosis can maintain that arrangement. Karyogamy in the basidium finally produces a diploid nucleus. Meiosis then reduces ploidy and recombines parental genetic material into haploid products. The timing of nuclear fusion, rather than simply the number of nuclei in a compartment, identifies the stages.

**Why not the others:** (A) calls the long dikaryotic phase diploid and incorrectly assigns reduction to mitosis. (C) makes plasmogamy nuclear fusion and ignores the stated paired nuclei. (D) adds separate nuclear chromosome sets as if they belonged to one tetraploid nucleus and reverses the effect of karyogamy.

**Key point:** Distinguish plasmogamy, a maintained dikaryon, karyogamy and meiosis.

---

**Q2 / A** Central partitioned versus peripheral ovule attachment

**Answer:** (A) Axile and parietal.

**Why:** The central placenta in X is associated with partitions that divide the ovary into locules. That combination identifies axile placentation. Y has ovules on the peripheral inner wall and no central placenta, identifying parietal placentation in the depicted simple case. Position of the placenta must be read together with whether the central axis is partitioned; an isolated central column would support a different identification.

**Why not the others:** (B) reverses central partitioned and peripheral attachment. (C) ignores X's septa, which distinguish it from free-central placentation, and Y does not show a single marginal suture. (D) would require ovules at the ovary base in X and a free-standing central column in Y, neither of which is shown.

**Key point:** Locate ovule attachment and then use septa to distinguish central placentation types.

---

**Q3 / C** Backbone hydrolysis distinguished from base-pair disruption

**Answer:** (C) The sugar-phosphate backbone is cut.

**Why:** Covalent phosphodiester linkages connect successive nucleotides along a DNA strand. Hydrolysing these bonds creates shorter backbone fragments and new ends. Hydrogen bonds between complementary bases are a different interaction and are not the enzyme's specified substrate. Fragmentation can eventually change duplex stability, but it does not mean the first chemical event was cleavage of base pairing.

**Why not the others:** (A) requires chemical conversion of nitrogenous bases, not backbone hydrolysis. (B) invokes peptide bonds, which join amino acids rather than DNA nucleotides. (D) changes the sugar's chemical identity and does not follow from cutting a phosphate linkage.

**Key point:** Separate covalent continuity within strands from complementary pairing between strands.

---

**Q4 / B** Gross fixation recovered from a net-exchange light curve

**Answer:** (B) 8 units and zero.

**Given:** Dark net exchange is -2 units; at L it is +6 units. Respiration is assumed constant.

**Formula/Principle:** Net fixation equals gross fixation minus respiration.

**Step 1:** Darkness gives respiratory release $R = 2$ units because gross photosynthetic fixation is zero.

**Step 2:** At L,

$$G = N + R = 6 + 2 = 8 \text{ units.}$$

**Step 3:** At the compensation point, $G = R$, so

$$N = G - R = 0.$$

**Why:** The graph records a difference between fixation and respiratory loss, not gross photosynthesis alone. The dark intercept supplies the missing respiratory term under the explicit constant-respiration assumption. At compensation the leaf can photosynthesise while gaining no net carbon over the interval.

**Why not the others:** (A) treats net uptake as gross and assigns respiration as net gain. (C) subtracts respiratory loss a second time. (D) calculates gross fixation correctly but counts the balancing respiratory flow as retained carbohydrate.

**Key point:** A zero net exchange need not mean either underlying flux is zero.

---

**Q5 / D** Countercurrent gradient preservation versus ADH permeability

**Answer:** (D) Less concentrated urine.

**Why:** ADH increases collecting-duct water permeability, permitting water to move down an osmotic gradient. It does not itself create the entire medullary concentration difference. Countercurrent exchange in the vasa recta helps preserve the gradient established by the loop and associated solute movements. Washing out that gradient reduces the driving force for water withdrawal even when permeability and ascending-limb transport remain available.

**Why not the others:** (A) confuses permeability regulation with creation of the external osmotic force. (B) ignores loss of the gradient despite one intact generating process. (C) concerns filtration in the renal corpuscle, which the stated medullary perturbation does not abolish.

**Key point:** Concentrated urine requires both water permeability and a maintained medullary gradient.

---

**Q6 / A** Unreduced parthenogenetic embryo with separately fertilised central cell

**Answer:** (A) 18 and 45.

**Given:** $n = 9$; the egg and each polar nucleus contain 18 chromosomes.

**Formula/Principle:** Count only nuclei that actually fuse; parthenogenesis excludes egg fertilisation.

**Step 1:** The unreduced egg develops without a sperm, so

$$N_{\text{embryo}} = 18.$$

**Step 2:** The two polar nuclei contribute $18 + 18$, while the normal sperm contributes 9:

$$N_{\text{endosperm}} = 18 + 18 + 9 = 45.$$

**Why:** The stem deliberately separates embryo formation from central-cell fertilisation. The ordinary diploid-embryo/triploid-endosperm shortcut presumes reduced female nuclei and fertilisation of the egg, neither of which applies here. Adding the nuclei specified for each event keeps maternal and paternal contributions distinct.

**Why not the others:** (B) fertilises a reduced egg and uses reduced polar nuclei. (C) handles the embryo correctly but replaces the stated unreduced polar nuclei by haploid ones. (D) doubles the egg complement despite the explicit absence of egg fusion.

**Key point:** Reconstruct ploidy from the actual participating nuclei when reproduction departs from the ordinary cycle.

---

**Q7 / C** Random union restores genotype proportions without changing allele frequency

**Answer:** (C) An increase to 0.48.

**Given:** Genotype frequencies are 0.50, 0.20, 0.30, with random gamete union and equal fertility.

**Formula/Principle:** Derive allele frequencies before applying Hardy-Weinberg proportions to the next generation.

**Step 1:**

$$p = 0.50 + \tfrac{1}{2}(0.20) = 0.60, \qquad q = 0.40.$$

**Step 2:** Random union gives

$$P(Aa) = 2pq = 2(0.60)(0.40) = 0.48.$$

**Step 3:** The increase is $0.48 - 0.20 = 0.28$, with unchanged allele frequencies.

**Why:** An initial population need not already have equilibrium genotype proportions. Random mating redistributes the conserved alleles into new genotype combinations. Thus genotype frequencies can change even when no allele is gained or lost; the stated assumptions justify the one-generation calculation.

**Why not the others:** (A) omits one of the two ordered gamete routes to a heterozygote. (B) uses $q^2$, the recessive-homozygote probability. (D) confuses conservation of allele frequency with conservation of genotype frequency.

**Key point:** Allele conservation and genotype equilibrium are different statements.

---

**Q8 / B** Conditional survival benefit separated from universal growth superiority

**Answer:** (B) The advantage can depend on antibiotic exposure.

**Why:** Survival under treatment demonstrates an advantage in that particular selective environment. Growth of both isolates after withdrawal shows that the drug is not a growth requirement for the survivors. The observations do not measure relative growth rates in all conditions or identify when resistance arose. They establish an environmental dependence of survival rather than a universal ranking of fitness.

**Why not the others:** (A) asserts a conversion of every sensitive cell, whereas many cells died and no mutation timing was measured. (C) contradicts resistant-isolate growth after removal. (D) generalises a treatment survival advantage to environments and growth measurements not tested.

**Key point:** Fitness claims must specify the environment and the measured component of success.

---

**Q9 / D** No-template contamination versus a true sample-specific PCR signal

**Answer:** (D) Contamination prevents a unique attribution.

**Why:** A no-template tube should not yield the target-specific amplification product when reagents and handling are free of contaminating amplifiable DNA. Its positive band exposes a source that could also contribute to the sample tube. The positive control shows the assay can amplify, but does not identify the source of each product. The sample may truly contain the target; this run cannot establish that independently of contamination.

**Why not the others:** (A) incorrectly turns an invalid positive inference into proof of absence. (B) conflicts with successful amplification in the controls. (C) substitutes an unsupported spontaneous sequence-creation mechanism for contaminating template.

**Key point:** Positive and negative controls answer different questions about an amplification result.

---

**Q10 / A** Producer and consumer production rankings under unequal transfer efficiency

**Answer:** (A) X has lower NPP without a necessary contradiction.

**Why:** Net primary production is gross production minus producer respiration, so the larger respiratory loss at fixed gross production lowers X's NPP. Secondary production measures new consumer biomass, not a fixed fraction of NPP in every real community. Differences in consumption, assimilation and consumer respiration can allow higher secondary production from a smaller NPP. Conservation still constrains each complete energy budget but does not demand identical efficiencies.

**Why not the others:** (B) reverses the producer budget using a consumer observation. (C) imposes an unstated equal-efficiency rule. (D) equates energy lost by producers to biomass retained by consumers, which are distinct fluxes.

**Key point:** Compare like production terms and avoid assuming universal trophic-transfer fractions.

---

**Q11 / C** A mortality-removal control challenges a survival-selection mechanism

**Answer:** (C) The same rise occurs without the proposed differential mortality.

**Why:** The proposed mechanism is a change in allele representation through unequal survival. A control in which no individuals are removed by mortality eliminates that route while retaining mating and the stated exclusions. An equal rise there requires another explanation or a faulty measurement, so it directly challenges the proposed cause of the frequency change. Heritability and prior variation are compatible with selection but do not by themselves establish that selection caused the measured shift.

**Why not the others:** (A) supplies a condition needed for inherited evolutionary change. (B) is expected when selection acts on existing variation. (D) shows an environment-dependent reversal, which supports rather than defeats conditional selection.

**Key point:** A causal test removes the proposed pathway while keeping relevant alternatives controlled.

---

**Q12 / D** Two-locus heterozygosity conditioned on a recessive lethal genotype

**Answer:** (D) 1/3.

**Given:** The loci assort independently; only aa is removed.

**Formula/Principle:** Condition the joint genotype probability on survival.

**Step 1:** Before selection,

$$P(AaBb) = \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{4}.$$

**Step 2:** Survival requires A-, so

$$P(\text{live}) = 1 - \tfrac{1}{4} = \tfrac{3}{4}.$$

**Step 3:** Every AaBb survives; therefore

$$P(AaBb \mid \text{live}) = \frac{1/4}{3/4} = \frac{1}{3}.$$

**Why:** The lethal genotype changes the denominator of the observed offspring distribution. Independence can be used before selection, and B does not affect survival. Conditioning excludes the dead aa classes instead of applying the unselected dihybrid ratio to live births.

**Why not the others:** (A) is the unconditioned conception probability. (B) multiplies survival by a single heterozygote frequency rather than conditioning. (C) uses the B-locus heterozygote probability alone and ignores the A-locus condition.

**Key point:** Identify the sampled surviving population before using Mendelian proportions.

---

**Q13 / A** Delayed opposing axon pulses and refractory collision

**Answer:** (A) They collide at 1.5 cm and 0.75 ms, then extinguish.

**Given:** $v = 20$ m/s $= 2$ cm/ms; P-Q separation is 2 cm and Q starts at 0.5 ms.

**Formula/Principle:** Meet the two travelling fronts, then test the refractoriness of the membrane behind each.

**Step 1:** At Q's stimulation time the P front is $2(0.5) = 1$ cm from P, so Q has not yet been excited.

**Step 2:** With t in ms and x in cm from P,

$$x_P = 2t, \qquad x_Q = 2 - 2(t - 0.5) = 3 - 2t.$$

**Step 3:** Equating positions gives

$$4t = 3, \quad t = 0.75 \text{ ms}, \quad x = 1.5 \text{ cm}.$$

**Step 4:** Beyond collision, either front would enter membrane just excited by the other. The elapsed times are below 2 ms, so that membrane is refractory and neither front propagates through.

**Why:** Action potentials require excitable membrane ahead of a regenerating front. They are not passive waves that simply superpose and re-emerge. The delayed stimulus can start at Q because the first wave has not reached it, but the subsequent collision puts recently excited membrane on both forward paths.

**Why not the others:** (B) treats regenerative excitation like linear wave superposition. (C) ignores the 1 ms travel time from P to Q. (D) ignores Q's delayed start and invents a speed increase.

**Key point:** Combine travel time with local recovery state when comparing successive neural excitations.

---

**Q14 / B** Equal initial osmolarity but different solute permeability and tonicity

**Answer:** (B) X preserves volume, while permeant Y can cause swelling.

**Why:** Equal initial osmolarities remove an initial water gradient but do not guarantee equal long-term tonic effects. Impermeant X remains outside and can balance the cell's retained internal solutes. Y enters the cell, adding intracellular osmotic particles while its own concentration tends to equalise across the membrane. Water then enters in response to the retained intracellular material, potentially exceeding the membrane's volume tolerance.

**Why not the others:** (A) ignores membrane permeability and equates initial osmolarity with sustained tonicity. (C) reverses the stated permeabilities and adds unsupported active transport. (D) assumes any external solute is hypertonic regardless of the matching initial total concentration.

**Key point:** Sustained cell-volume effects depend on permeability as well as total osmolarity.

---

**Q15 / C** Asymmetric insert orientation inferred from a two-site circular digest

**Answer:** (C) 2 kb and 6 kb.

**Given:** The backbone short arc E-to-H is 1 kb; the internal H is 1 kb from the adjacent insert junction in orientation X.

**Formula/Principle:** Two restriction sites on circular DNA delimit two complementary arcs.

**Step 1:** The short arc from backbone H through the nearby E junction to insert H is

$$1 + 1 = 2 \text{ kb}.$$

**Step 2:** The complementary arc is

$$8 - 2 = 6 \text{ kb}.$$

**Step 3:** Reversing the insert gives $1 + 2 = 3$ kb and $8 - 3 = 5$ kb, providing an orientation contrast.

**Why:** Total insert size alone cannot distinguish an orientation. An asymmetric internal site makes one distance to a fixed backbone site orientation-dependent. Complete digestion cuts both sites, so the two fragment lengths must sum to the entire recombinant length.

**Why not the others:** (A) uses the backbone arc without adding the insert segment. (B) lists lengths that do not account for the 8 kb molecule. (D) is the predicted pair for the reversed insert, not X.

**Key point:** Use an asymmetric insert site and a fixed backbone site to infer orientation from a circular digest.

---

## Chemistry

**Q16 / C** Geometrical isomerism changes vector addition of identical bond dipoles

**Answer:** (C) $\sqrt{3}d$ and zero.

**Given:** The cis C-F vectors point at 60 and 120 degrees to the rightward horizontal; trans vectors are opposite.

**Formula/Principle:** Add dipole vectors, rather than bond-polarity magnitudes alone.

**Step 1:** For cis, horizontal components cancel and vertical components add:

$$\mu_x = d\cos 60^\circ + d\cos 120^\circ = 0, \quad \mu_y = 2d\sin 60^\circ = \sqrt{3}d.$$

**Step 2:** For trans,

$$\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2 = 0.$$

**Why:** Both isomers contain the same polar bonds, but their fixed relative directions differ. The idealised model isolates vector geometry from other bond contributions; it predicts a magnitude ratio within that model, not a measured molecular dipole in unspecified conditions. Cis vectors are separated by 60 degrees and cannot cancel.

**Why not the others:** (A) keeps only an incorrect single component. (B) adds magnitudes as parallel vectors and then leaves a residual for opposite vectors. (D) reverses the cancellation geometry of the isomers.

**Key point:** Use the angle between bond vectors when translating bond polarity into molecular polarity.

---

**Q17 / D** d8 square-planar versus tetrahedral pairing at equal coordination number

**Answer:** (D) X is diamagnetic and Y has two unpaired electrons.

**Why:** Neutral nickel loses its 4s electrons to give the Ni(II) d8 configuration. The strong-field cyanide square-planar complex pairs the d electrons and uses the usual dsp2 description, leaving none unpaired. The tetrahedral chloride complex uses the usual sp3 description without the corresponding d-electron pairing, leaving two unpaired electrons. Equal coordination number therefore does not establish equal geometry or magnetic behaviour.

**Why not the others:** (A) reverses the pairing assignments. (B) treats coordination number as a sufficient electron-pairing rule. (C) gives the wrong d count and hence the wrong magnetic premise.

**Key point:** Determine oxidation state and d count, then use ligand and geometry information to assess pairing.

---

**Q18 / A** Standard versus composition-dependent spontaneity on opposite sides of K

**Answer:** (A) Forward is favourable at Q1 and unfavourable at Q2.

**Given:** Standard changes, T=300 K, RT=2.5 kJ/mol and two specified dimensionless quotients.

**Formula/Principle:** $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, while $\Delta G = \Delta G^\circ + RT\ln Q$.

**Step 1:** Convert entropy to kJ units:

$$\Delta G^\circ = 20 - 300(0.050) = 5 \text{ kJ/mol}.$$

**Step 2:** For Q1,

$$\Delta G_1 = 5 + 2.5(-4) = -5 \text{ kJ/mol}.$$

**Step 3:** For Q2,

$$\Delta G_2 = 5 + 2.5(-1) = +2.5 \text{ kJ/mol}.$$

**Step 4:** The equilibrium quotient is $K = e^{-5/2.5} = e^{-2}$. Thus Q1 is below K and Q2 above K, independently checking the two directions.

**Why:** Standard-state spontaneity concerns Q=1, not every possible composition. A sufficiently small product-to-reactant quotient can make a positive standard Gibbs change compatible with a negative actual Gibbs change. At fixed temperature, K is fixed but Q can still distinguish mixtures on opposite sides of equilibrium.

**Why not the others:** (B) uses the standard change for nonstandard mixtures. (C) equates fixed temperature with equilibrium at every composition. (D) reverses the sign of the quotient correction.

**Key point:** Evaluate the actual Gibbs change or compare Q with K before assigning reaction direction.

---

**Q19 / B** Intramolecular aldol ring closure and surviving donor aldehyde

**Answer:** (B) A five-membered ring with an aldehyde substituent.

**Why:** The new bond joins C2 to C6. The closed path is C2-C3-C4-C5-C6, containing five carbon atoms. C6's carbonyl becomes an alcohol after addition and protonation, while C1 remains an aldehyde attached to C2. The donor's carbonyl is not consumed when its adjacent enolate carbon makes the new bond. Stopping before dehydration preserves the aldol alcohol group.

**Why not the others:** (A) counts C1 as part of a ring it does not join and removes the surviving aldehyde. (C) omits one of the atoms on the closed path and changes the retained functional group. (D) retains the attacked carbonyl even though nucleophilic addition converts it to an alcohol.

**Key point:** Trace the newly closed bond path and track donor and acceptor carbonyls separately.

---

**Q20 / C** Active-anode replacement of cathodically removed metal ions

**Answer:** (C) Copper transfers from anode to cathode without net bulk Cu2+ change.

**Why:** Reduction at the cathode consumes one Cu2+ for each copper atom deposited. Oxidation at the copper anode supplies one Cu2+ for each copper atom dissolved. The same charge passes through both electrodes and the specified reaction transfers two electrons per atom at each, so the molar and mass changes cancel in solution. This conclusion depends on an active copper anode, rather than an inert anode that might oxidise water.

**Why not the others:** (A) assigns reduction to both electrodes and neglects anodic dissolution. (B) reverses the cathodic reaction. (D) omits the reduction required to carry the stated current.

**Key point:** Electrode material can determine whether electrolysis changes solution concentration or merely transfers metal.

---

**Q21 / B** Carbon loss in Hofmann degradation followed by an amine-class test

**Answer:** (B) Aniline forms and gives the carbylamine reaction.

**Why:** Hofmann bromamide degradation converts a primary amide to a primary amine with one fewer carbon atom. In benzamide the carbonyl carbon is removed, leaving the phenyl group directly attached to nitrogen as aniline. Aniline is an aromatic primary amine and can give the carbylamine reaction with chloroform and alcoholic alkali. The two stages therefore test carbon accounting and amine class rather than merely retention of nitrogen.

**Why not the others:** (A) retains the carbonyl carbon as a methylene, which is characteristic of a different reduction route. (C) substitutes amide hydrolysis for Hofmann degradation and assigns an amine test to a carboxylic acid. (D) adds a methyl group and makes a secondary amine, which does not give the ordinary carbylamine test.

**Key point:** Hofmann degradation removes the amide carbonyl carbon while preserving a primary amine product.

---

**Q22 / D** Residual oxidant tracked through two electron-equivalent conversions

**Answer:** (D) 0.060 mol.

**Given:** Fe2+ loses one electron per ion, dichromate accepts six, and iodine consumes two thiosulfates per molecule.

**Formula/Principle:** Follow electron equivalents through the residual oxidant, not the original dose alone.

**Step 1:** Dichromate consumed by iron is

$$n_{\text{used}} = 0.120/6 = 0.020 \text{ mol}.$$

**Step 2:** Residual dichromate is

$$0.030 - 0.020 = 0.010 \text{ mol}.$$

**Step 3:** Oxidising two iodides to one iodine releases two electrons. Therefore the residual accepts enough electrons to form

$$n(\text{I}_2) = 6(0.010)/2 = 0.030 \text{ mol}.$$

**Step 4:**

$$n(\text{S}_2\text{O}_3^{2-}) = 2(0.030) = 0.060 \text{ mol}.$$

**Why:** The back-titration measures oxidant left after the initial iron reaction. Electron conservation links dichromate to iodine even though their molecular coefficients differ. The supplied iodine-thiosulfate equation then determines the final reagent count.

**Why not the others:** (A) reports the dichromate consumed by iron. (B) stops at the iodine amount. (C) titrates the full original dichromate dose without subtracting iron consumption.

**Key point:** Resolve the first reaction before interpreting a residual-reagent titration.

---

**Q23 / A** Reverse temperature sensitivity reconstructed from reaction enthalpy

**Answer:** (A) 3.

**Given:** $E_{a,f} = 40$ kJ/mol and $\Delta H = -20$ kJ/mol for the forward reaction.

**Formula/Principle:** For the same transition state, $E_{a,f} - E_{a,r} = \Delta H$; an Arrhenius log-ratio is proportional to activation energy over the same temperature interval.

**Step 1:**

$$E_{a,r} = 40 - (-20) = 60 \text{ kJ/mol}.$$

**Step 2:** For either direction,

$$\ln \frac{k(T_2)}{k(T_1)} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right).$$

**Step 3:** Divide the reverse expression by the forward one:

$$\ln \frac{k_r(T_2)}{k_r(T_1)} = 2\frac{60}{40} = 3.$$

**Why:** An exothermic forward reaction leaves products lower in energy, so the reverse direction must climb a larger barrier to the same transition state. The unknown temperatures cancel only because both comparisons use the same interval and the stated Arrhenius assumptions.

**Why not the others:** (B) subtracts the reaction enthalpy magnitude from the forward barrier. (C) assumes equal barriers for an exothermic pair. (D) doubles the measured log-ratio without using the actual barrier ratio.

**Key point:** Relate forward and reverse barriers before comparing their temperature sensitivities.

---

**Q24 / C** Conserved acyl skeleton across ester aminolysis and amide hydrolysis

**Answer:** (C) Benzamide followed by benzoic acid.

**Why:** Ammonia replaces the ester's methoxy group at the acyl centre to form benzamide, preserving the phenyl-carbonyl carbon framework. Acidic hydrolysis of the amide then replaces the amino group by hydroxyl at that same acyl carbon, giving benzoic acid. Neither stage is the carbon-removing Hofmann reaction or a carbonyl reduction. The methyl group of the ester leaves with the alcohol-derived fragment rather than migrating onto nitrogen.

**Why not the others:** (A) removes the carbonyl carbon and then nitrogen without either required reaction. (B) reduces the carbonyl to methylene, which the stated reagents do not do. (D) changes the acyl skeleton from benzoyl to acetyl and disconnects the phenyl group.

**Key point:** In ordinary ester-amide-acid conversions, track the conserved acyl carbon separately from the leaving group.

---

**Q25 / D** Proton relocation distinguishes tautomerism from resonance

**Answer:** (D) They are tautomers.

**Why:** The aldehyde and enol forms preserve the molecular formula but differ in which atom bears a hydrogen and where the multiple bond lies. Their interconversion involves nuclear motion and bond reorganisation, so they represent different constitutional forms in a tautomeric equilibrium. Resonance descriptions instead retain the positions and connectivity of nuclei while redistributing electrons.

**Why not the others:** (A) ignores the hydrogen transfer. (B) confuses a bond-order change with a change in atom counts. (C) requires the same double-bond connectivity with different spatial arrangements, which is not the relationship here.

**Key point:** Moving a proton distinguishes tautomerism from electron-only resonance bookkeeping.

---

**Q26 / B** A reverse-osmosis pressure limit fixes the removable solvent fraction

**Answer:** (B) 2/3.

**Given:** Solute-particle amount and temperature stay fixed; the applied pressure difference is three times the initial osmotic pressure.

**Formula/Principle:** Reverse osmosis stops when the growing osmotic pressure equals the imposed pressure difference; ideal osmotic pressure is inversely proportional to solution volume at fixed particle amount.

**Step 1:**

$$\pi(V) = n_{\text{particles}} RT/V = \pi_0 V_0/V.$$

**Step 2:** At the limit,

$$3\pi_0 = \pi_0 V_0/V_f \Rightarrow V_f = V_0/3.$$

**Step 3:** The removed fraction is

$$(V_0 - V_f)/V_0 = 1 - 1/3 = 2/3.$$

**Why:** Removing solvent concentrates the retained salt, which increases the pressure needed to continue forcing water out. A fixed applied pressure therefore cannot remove all the water in this ideal limiting model. The salt dissociation factor is already contained in pi0 and cancels in the volume ratio.

**Why not the others:** (A) gives the fraction remaining. (C) would require a final osmotic pressure four times the initial value. (D) would balance only twice the initial osmotic pressure.

**Key point:** In reverse osmosis at fixed pressure, update the retained solution concentration as solvent leaves.

---

**Q27 / A** Isotope abundance from a weighted atomic mass followed by sample mole accounting

**Answer:** (A) 0.0150 mol.

**Given:** The two isotope masses are 63 u and 65 u; their weighted mean is 63.6 u. The sample contains 0.0500 mol atoms.

**Formula/Principle:** Mean atomic mass is weighted by the number fraction of each isotope, which is also its mole fraction.

**Step 1:** If f is the heavier-isotope fraction,

$$63(1-f)+65f=63.6 \Rightarrow 2f=0.6 \Rightarrow f=0.300.$$

**Step 2:**

$$n_{65}=fn_{\text{total}}=0.300(0.0500)=0.0150 \text{ mol}.$$

**Step 3:** The lighter isotope contributes 0.0350 mol, and the sum returns the stated total.

**Why:** Isotopes of an element have different masses but contribute to the same atomic population. The mean is therefore a number-weighted average, not an equal-weight midpoint unless the abundances are equal. Multiplying the inferred fraction by total atom moles gives isotope moles directly, without treating the average mass as the mass of every atom.

**Why not the others:** (B) gives the lighter-isotope amount. (C) treats the 0.6 u shift above 63 u as a fraction without dividing by the 2 u isotope gap. (D) assumes equal abundance despite the mean lying below 64 u.

**Key point:** Infer isotope number fractions from the full mass gap before converting a total sample amount.

---

**Q28 / D** An electron-gain trend challenged by the full aqueous formation cycle

**Answer:** (D) F is favoured by 160 kJ per mole of atoms.

**Given:** Each entry is on the same per-mole-of-X-atoms basis.

**Formula/Principle:** Add the successive atomisation, electron-gain and hydration enthalpies.

**Step 1:**

$$\Delta H_F=80-330-500=-750 \text{ kJ/mol}.$$

**Step 2:**

$$\Delta H_{Cl}=120-350-360=-590 \text{ kJ/mol}.$$

**Step 3:**

$$\Delta H_F-\Delta H_{Cl}=-160 \text{ kJ/mol}.$$

**Why:** Isolated-atom electron gain is only one stage in forming aqueous halide from a molecular gas. More favourable hydration and lower atomisation cost outweigh chlorine's 20 kJ advantage in the electron-gain step. This is an enthalpy-cycle conclusion from the supplied data, not by itself a complete Gibbs-energy calculation of electrode potential.

**Why not the others:** (A) both selects the wrong overall preference and misuses a single-stage comparison. (B) gives the isolated electron-gain difference rather than the net difference. (C) equates electron count with equal energetics.

**Key point:** A periodic trend for one gas-phase step need not determine a multistage aqueous process.

---

**Q29 / C** Triprotic amino-acid charge and electrode direction

**Answer:** (C) Net -1, towards the anode.

**Why:** At pH 6 both carboxyl groups are predominantly deprotonated, contributing two negative charges. The ammonium group remains predominantly protonated below its pKa of 9, contributing one positive charge. Their sum is -1, so the ion moves towards the positive electrode, the anode. The additional acidic side group makes the neutral-side-chain shortcut inapplicable.

**Why not the others:** (A) cancels only one carboxyl charge and overlooks the second. (B) treats carboxyl groups as neutral despite pH exceeding their pKa values. (D) removes the ammonium proton prematurely and also reverses the direction for a negative ion.

**Key point:** Count every ionisable group at the stated pH before inferring migration.

---

**Q30 / A** Two buffer observations followed by equivalence-point hydrolysis

**Answer:** (A) 8.50.

**Given:** Equal acid amounts; two partial-neutralisation buffers and one equivalence-point salt solution.

**Formula/Principle:** In a buffer, $\mathrm{pH} = \mathrm{p}K_a + \log[b/(n-b)]$; at equivalence use conjugate-base hydrolysis.

**Step 1:** Subtract the two pH equations. Their difference is $\log 9$, giving

$$\frac{15/(n-15)}{5/(n-5)} = 9 \;\Rightarrow\; 3(n-5) = 9(n-15) \;\Rightarrow\; n = 20 \text{ mmol}.$$

**Step 2:** For the first buffer,

$$4 - \log 3 = \mathrm{p}K_a + \log(5/15) = \mathrm{p}K_a - \log 3,$$

hence $\mathrm{p}K_a = 4$ and $K_a = 10^{-4}$.

**Step 3:** At equivalence,

$$C_{A^-} = 0.020/0.200 = 0.100 \text{ M}, \quad K_b = K_w/K_a = 10^{-10}.$$

**Step 4:**

$$[\mathrm{OH}^-] \simeq \sqrt{K_b C} = \sqrt{10^{-11}} = 10^{-5.5} \text{ M}, \quad \mathrm{pH} = 14 - 5.5 = 8.50.$$

**Step 5:** Hydrolysed fraction is about $3.2 \times 10^{-5}$, supporting the small-hydrolysis approximation.

**Why:** The two buffer observations jointly identify both total acid and its dissociation constant. Exact neutralisation removes the HA/A- buffer and creates a basic salt, so the equivalence pH must be found from hydrolysis rather than extending the buffer formula to a zero denominator.

**Why not the others:** (B) uses pKa as though every titration state were half-neutralised. (C) assumes a weak-acid/strong-base equivalence is neutral. (D) overestimates hydroxide by using an incorrect concentration or hydrolysis exponent.

**Key point:** Infer buffer parameters from partial neutralisation, then change the equilibrium model at equivalence.

---

## Mathematics

**Q31 / A** Centre-line and chord constraints precede an external tangent-length calculation

**Answer:** (A) $\sqrt{2}$.

**Given:** Write the centre as $(h, h)$, since it lies on y=x.

**Formula/Principle:** Equal centre-to-endpoint distances determine h; an external point has squared tangent length equal to its squared centre distance minus radius squared.

**Step 1:** Equidistance from A and B gives

$$h^2 + (h-1)^2 = (h-2)^2 + (h-3)^2 \Rightarrow 8h = 12 \Rightarrow h = 3/2.$$

**Step 2:** The radius squared is

$$r^2 = (3/2)^2 + (3/2 - 1)^2 = 5/2.$$

**Step 3:** The origin-to-centre distance squared is

$$OC^2 = (3/2)^2 + (3/2)^2 = 9/2 > r^2,$$

so a real tangent exists.

**Step 4:**

$$OT = \sqrt{OC^2 - r^2} = \sqrt{9/2 - 5/2} = \sqrt{2}.$$

**Worked derivation:**

$$CT \perp OT, \qquad OT^2 + CT^2 = OC^2, \qquad OT = \sqrt{2}.$$

**Why:** The line containing the centre must be combined with the perpendicular-bisector condition supplied by the two points. The tangent is drawn from a different point, the origin, and its length is not the radius or the centre distance. The positive power check confirms that the proposed external tangent is geometrically possible before taking its square root.

**Why not the others:** (B) reports the squared tangent length as a length. (C) gives the circle radius. (D) gives the origin-to-centre distance.

**Key point:** Recover the centre from chord information before using the right triangle at a tangent contact.

---

**Q32 / D** Ellipse reconstructed from chord midpoint, direction and length

**Answer:** (D) 27/2.

**Given:** The half-chord has length $\sqrt{5}$ and direction proportional to $(1, -2)$.

**Formula/Principle:** Recover endpoints from midpoint and displacement, then impose the ellipse equation at each.

**Step 1:** Since $|(1, -2)| = \sqrt{5}$, the endpoints are

$$(1, 1) \pm (1, -2) = (2, -1), (0, 3).$$

**Step 2:** Substituting $(0, 3)$ gives

$$9/b^2 = 1 \;\Rightarrow\; b^2 = 9.$$

**Step 3:** Substituting $(2, -1)$ gives

$$4/a^2 + 1/9 = 1 \;\Rightarrow\; a^2 = 9/2.$$

**Step 4:**

$$a^2 + b^2 = 9/2 + 9 = 27/2.$$

**Worked derivation:**

$$\frac{2^2}{9/2} + \frac{(-1)^2}{9} = 1, \qquad \frac{0^2}{9/2} + \frac{3^2}{9} = 1.$$

**Why:** Chord length alone does not specify its endpoints. Slope fixes the direction and midpoint fixes its placement. Once both endpoint constraints are used, the two positive semiaxis squares

are determined. The symbols a and b are not assumed to identify the major axis; imposing a>b would wrongly reject the actual ellipse.

**Why not the others:** (A) gives only b squared. (B) gives only a squared. (C) treats both semiaxes as b by ignoring the off-axis endpoint.

**Key point:** Reconstruct the actual chord before solving for an unknown conic.

---

**Q33 / B** Total enclosed area with both a touching point and absolute-value boundary changes

**Answer:** (B) $8(\sqrt{2}-1)/3$.

**Given:** The lower curve changes expression at $x = \pm 1$.

**Formula/Principle:** Find intersections and integrate upper minus lower on each sign region.

**Step 1:** $|x^2 - 1| \le 1$ gives $-\sqrt{2} \le x \le \sqrt{2}$. The curves also touch at x=0.

**Step 2:** By evenness,

$$A = 2\left[\int_0^1 x^2\,dx + \int_1^{\sqrt{2}} (2 - x^2)\,dx\right].$$

**Step 3:**

$$A = 2\left[\frac{1}{3} + \left(2x - \frac{x^3}{3}\right)_1^{\sqrt{2}}\right] = \frac{8(\sqrt{2}-1)}{3}.$$

**Worked derivation:**

$$A = 2\left[\frac{1}{3} + \frac{4\sqrt{2}}{3} - \frac{5}{3}\right] = \frac{8\sqrt{2}-8}{3}.$$

**Why:** The absolute value creates two different polynomial gaps. The central touching point does not eliminate either adjacent region's area, while symmetry lets one integrate only the nonnegative half. Using a single polynomial over the whole interval would integrate the wrong lower boundary on one part.

**Why not the others:** (A) drops the finite contribution at the split point. (C) computes only the right-hand half. (D) counts only the central interval and omits the outer pieces.

**Key point:** Determine both the feasible interval and the internal sign changes of an absolute-value boundary.

---

**Q34 / C** A squared-sine substitution transforms a quartic trigonometric denominator

**Answer:** (C) $\pi/8$.

**Given:** The numerator contains the derivative of $\sin^2 x$ up to a factor.

**Formula/Principle:** Substitute $u = \sin^2 x$ and transform both limits.

**Step 1:**

$$du = 2\sin x \cos x\,dx, \qquad x = 0 \Rightarrow u = 0, \quad x = \pi/2 \Rightarrow u = 1.$$

**Step 2:**

$$I = \frac{1}{2}\int_0^1 \frac{du}{1+u^2} = \frac{1}{2}[\arctan u]_0^1 = \frac{\pi}{8}.$$

**Worked derivation:**

$$I = \tfrac{1}{2}(\arctan 1 - \arctan 0) = \tfrac{1}{2} \cdot \tfrac{\pi}{4}.$$

**Why:** Squaring sine absorbs the paired sine-cosine factor and turns the fourth power into a quadratic in the new variable. The factor one-half is required by the derivative; the transformed limits remove any need to back-substitute.

**Why not the others:** (A) misses the factor one-half. (B) replaces the varying denominator by a constant. (D) ignores both the denominator and the substitution factor.

**Key point:** Match a composite power with its differential before choosing an integration formula.

---

**Q35 / A** Inverse curvature obtained by twice differentiating a composition

**Answer:** (A) 1/8.

**Given:** $f(1) = 1$, and $f'(x) = 1 + 1/x > 0$ on its domain.

**Formula/Principle:** Differentiate $f(g(y)) = y$ twice; the second derivative of an inverse is not the reciprocal of the original second derivative.

**Step 1:**

$$f'(g(y))g'(y) = 1, \qquad g'(1) = 1/f'(1) = 1/2.$$

**Step 2:**

$$f''(g(y))(g'(y))^2 + f'(g(y))g''(y) = 0.$$

**Step 3:** Since $f''(1) = -1$,

$$-1(1/2)^2 + 2g''(1) = 0 \;\Rightarrow\; g''(1) = 1/8.$$

**Worked derivation:**

$$g''(1) = -\frac{f''(1)}{(f'(1))^3} = -\frac{-1}{2^3} = \frac{1}{8}.$$

**Why:** Monotonicity establishes an inverse with a nonzero derivative at the relevant point. Both chain-rule factors matter when differentiating again because the original independent variable is itself g(y). Checking f(1)=1 identifies the correct preimage before evaluating derivatives.

**Why not the others:** (B) loses the minus sign from solving the twice-differentiated identity. (C) omits one factor of f prime. (D) combines that missing factor with the sign error.

**Key point:** Derive inverse-function curvature from the composition identity, including every chain-rule factor.

---

**Q36 / B** A reconstructed rational involution without real fixed points

**Answer:** (B) It is its own inverse and has no real fixed point.

**Given:** Both specified inputs lie in the natural domain.

**Formula/Principle:** Recover parameters, check the composite with domain exclusions, and solve f(x)=x separately.

**Step 1:**

$$-b/a = 2 \Rightarrow b = -2a, \quad -a/(1-a) = 3 \Rightarrow a = 3/2, \quad b = -3.$$

**Step 2:** Thus

$$f(x) = \frac{3x-6}{2x-3}, \qquad x \neq 3/2.$$

Solving y=f(x) for x gives the same expression with y, and f(x) never equals 3/2.

**Step 3:** A fixed point would satisfy

$$2x^2 - 6x + 6 = 0, \quad \Delta = 36 - 48 = -12 < 0.$$

**Worked derivation:**

$$f(f(x)) = x \quad (x \neq 3/2), \qquad f^{-1}(y) = \frac{3y-6}{2y-3}.$$

**Why:** Self-inverse behaviour concerns applying the function twice, whereas a fixed point concerns applying it once. A real involution need not have a real fixed point when its domain omits a separating value. Retaining the pole and excluded range value makes the inverse statement precise.

**Why not the others:** (A) contradicts the negative fixed-point discriminant and the explicit inverse. (C) assumes self-inverse implies real fixed points. (D) ignores that the numerator is not proportional to the denominator.

**Key point:** Verify inverse, domain and fixed-point claims independently.

---

**Q37 / D** Cyclic matrix construction, determinant factorisation and inverse-trace parameters

**Answer:** (D) -1.

**Given:** I is the 3 by 3 identity and A is the stated cyclic permutation matrix.

**Formula/Principle:** Construct Bt, compute its determinant and diagonal cofactors, then impose the inverse-trace constraint only where the determinant is nonzero.

**Step 1:**

$$B_t = \begin{pmatrix} 1 & 1 & t \\ t & 1 & 1 \\ 1 & t & 1 \end{pmatrix}.$$

**Step 2:** Expansion gives

$$\det B_t = t^3 - 3t + 2 = (t-1)^2(t+2).$$

Therefore $t \neq 1, -2$.

**Step 3:** Each diagonal cofactor is $1 - t$, so

$$\mathrm{tr}(B_t^{-1}) = \frac{3(1-t)}{(t-1)^2(t+2)} = -\frac{3}{(t-1)(t+2)}.$$

**Step 4:** The required trace gives

$$(t-1)(t+2) = 4 \Rightarrow t^2 + t - 6 = 0 \Rightarrow t = 2, -3.$$

Both are admissible and their sum is -1.

**Worked derivation:**

$$\det B_2 = 4, \quad \det B_{-3} = -16, \quad \mathrm{tr}(B_2^{-1}) = \mathrm{tr}(B_{-3}^{-1}) = -3/4.$$

**Why:** The trace of an inverse is obtained from diagonal cofactors divided by the determinant, not from reciprocal diagonal entries. Cancelling a factor does not restore the singular parameter values. The resulting quadratic admits two different invertible matrices with the same inverse trace.

**Why not the others:** (A) reverses the sign of the root sum. (B) reports a singular value that must be excluded. (C) retains only the positive admissible parameter.

**Key point:** Keep singular exclusions when reducing a parameterised inverse condition.

---

**Q38 / A** A conditional subset count grouped by fixed range

**Answer:** (A) 1/3.

**Given:** Unordered three-ticket subsets are equally likely; their range is fixed at 4.

**Formula/Principle:** Count the conditional sample space by its minimum and then count the subsets containing 4.

**Step 1:** The minimum m can be 1,2,3, and the maximum is m+4. The third ticket has three choices strictly between them, giving

$$N = 3(3) = 9.$$

**Step 2:** For each minimum, exactly one allowed interior label is 4. The favourable sets are

$$\{1,4,5\}, \quad \{2,4,6\}, \quad \{3,4,7\}.$$

**Step 3:**

$$P = 3/9 = 1/3.$$

**Worked derivation:**

$$P(4 \text{ selected} \mid \text{range } 4) = \frac{\sum_{m=1}^{3} 1}{\sum_{m=1}^{3} 3} = \frac{1}{3}.$$

**Why:** The range condition changes the sample space from all three-element subsets to nine constrained subsets. Minimum-based grouping counts each once. The label 4 is always an interior candidate in these groups, never their endpoint, which makes the numerator transparent.

**Why not the others:** (B) uses the unconditional inclusion probability. (C) counts only one favourable set. (D) reports the complement.

**Key point:** Organise a range-conditioned selection by its minimum and maximum.

---

**Q39 / C** Tangent-cotangent sum with denominator and double-angle branch checks

**Answer:** (C) $\pi/2$.

**Given:** Both tangent and cotangent must exist.

**Formula/Principle:** Combine the fractions before solving the resulting double-angle equation.

**Step 1:**

$$\tan x + \cot x = \frac{1}{\sin x \cos x} = 4 \Rightarrow \sin 2x = 1/2.$$

**Step 2:** In $0 < 2x < 2\pi$, the solutions are $2x = \pi/6, 5\pi/6$, so

$$x = \pi/12, 5\pi/12.$$

**Step 3:** Both satisfy the original nonzero denominators, and their sum is $\pi/2$.

**Worked derivation:**

$$\frac{\pi}{12} + \frac{5\pi}{12} = \frac{\pi}{2}.$$

**Why:** Combining tangent and cotangent uses their common denominator and exposes the sign condition. In the second quadrant of x, the product sine-cosine is negative, so no further positive-right-side solutions occur. The excluded point pi/2 cannot be introduced by a transformed equation.

**Why not the others:** (A) adds reflected points that do not satisfy the sign condition. (B) halves the required sum. (D) includes inadmissible periodic branches.

**Key point:** Preserve the original denominators while using a trigonometric identity.

---

**Q40 / D** A plane-pencil endpoint selected by perpendicularity before a distance calculation

**Answer:** (D) $2\sqrt{2}$.

**Given:** The required plane belongs to the pencil through the two-plane intersection.

**Formula/Principle:** Use a complete homogeneous pencil and perpendicular normal vectors, then the point-plane distance.

**Step 1:** Write

$$\alpha(x + y - 1) + \beta(y + z - 1) = 0,$$

where alpha and beta are not both zero. Its normal is $(\alpha, \alpha + \beta, \beta)$.

**Step 2:** Perpendicularity gives

$$(\alpha, \alpha + \beta, \beta) \cdot (2, -1, 1) = \alpha = 0.$$

Thus beta is nonzero and the plane is $y + z - 1 = 0$.

**Step 3:**

$$d = \frac{|1 + 4 - 1|}{\sqrt{0^2 + 1^2 + 1^2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}.$$

**Worked derivation:**

$$\mathbf{n} = (0, 1, 1), \quad \mathbf{n} \cdot (2, -1, 1) = 0, \quad d = 2\sqrt{2}.$$

**Why:** The desired plane is exactly one of the original planes. A pencil written with a fixed nonzero coefficient of the first plane can accidentally omit it, so both coefficients are retained until perpendicularity is imposed. The distance must use the recovered plane rather than either an assumed midpoint plane or the given perpendicular plane.

**Why not the others:** (A) halves the correct distance. (B) omits normal-vector normalisation. (C) substitutes an incorrect constant in the plane equation.

**Key point:** A complete plane pencil includes either original plane as a possible answer.

---

**Q41 / B** Reciprocal projection constraints determine a scale-free length ratio

**Answer:** (B) $\sqrt{7/3}$.

**Given:** Both relations describe vector projections, not only scalar lengths.

**Formula/Principle:** $\text{proj}_{\mathbf{v}}\,\mathbf{u} = (\mathbf{u}\cdot\mathbf{v})\mathbf{v}/|\mathbf{v}|^2$.

**Step 1:** The first relation gives $\mathbf{a}\cdot\mathbf{b} = |\mathbf{b}|^2$; the second gives $\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}|^2/4$. Hence $|\mathbf{a}|^2 = 4|\mathbf{b}|^2$.

**Step 2:** Let $s = |\mathbf{b}|^2 > 0$. Then

$$|\mathbf{a}+\mathbf{b}|^2 = 4s + s + 2s = 7s, \quad |\mathbf{a}-\mathbf{b}|^2 = 4s + s - 2s = 3s.$$

**Step 3:** Take the positive square root of the ratio.

**Worked derivation:**

$$\frac{|\mathbf{a}+\mathbf{b}|}{|\mathbf{a}-\mathbf{b}|} = \sqrt{\frac{7s}{3s}} = \sqrt{\frac{7}{3}}.$$

**Why:** The projections determine both a norm ratio and the mutual dot product. Neither vector needs a coordinate representation. Positivity of s verifies that the denominator is nonzero and that taking the length ratio is valid.

**Why not the others:** (A) uses the wrong dot-product contribution. (C) reports the squared ratio. (D) assumes the plus and minus combinations have equal length, which would require orthogonality.

**Key point:** Translate two projection statements into dot-product and norm constraints before expanding lengths.

---

**Q42 / D** A unit-circle distance product fixes a higher power without fixing the point

**Answer:** (D) -1.

**Given:** z lies on the unit circle and the product of its distances from 1 and -1 is one.

**Formula/Principle:** Multiply complex moduli and use the unit-circle relation before taking powers.

**Step 1:**

$$|z^2 - 1| = 1.$$

Put $w = z^2$, so $|w| = 1$.

**Step 2:**

$$1 = |w-1|^2 = |w|^2 + 1 - 2\,\mathrm{Re}\,w = 2 - 2\,\mathrm{Re}\,w.$$

Thus $\mathrm{Re}\,w = 1/2$.

**Step 3:** The unit-circle possibilities are

$$w = \tfrac{1}{2} \pm i\tfrac{\sqrt{3}}{2} = e^{\pm i\pi/3},$$

giving $w^3 = -1$.

**Worked derivation:**

$$z^6 = (z^2)^3 = w^3 = e^{\pm i\pi} = -1.$$

**Why:** The distance product becomes a single modulus after factoring z squared minus one. Both conjugate unit-circle solutions yield the same requested power, so selecting one sign is unnecessary and would risk losing admissible z values.

**Why not the others:** (A) uses a full-turn instead of a half-turn for w cubed. (B) assigns a quarter-turn not implied by the distance constraint. (C) makes the opposite unsupported quarter-turn choice.

**Key point:** A transformed complex variable can retain multiple values while a requested power is unique.

---

**Q43 / C** Positive-root and reciprocal-gap constraints on an integer parameter

**Answer:** (C) 3.

**Given:** k is an integer in the stated finite range.

**Formula/Principle:** Use discriminant and Vieta's relations, then express the reciprocal gap in coefficients.

**Step 1:**

$$D = k^2 - 4k - 12 = (k-6)(k+2).$$

In the allowed range, distinct real roots require k>6. Their sum k and product k+3 are then positive, so both roots are positive.

**Step 2:**

$$|1/r - 1/s| = \frac{|r-s|}{rs} = \frac{\sqrt{D}}{k+3} > \frac{1}{2}.$$

**Step 3:** Squaring positive sides gives

$$4D > (k+3)^2 \;\Rightarrow\; 3k^2 - 22k - 57 > 0.$$

**Step 4:** The upper root of this last quadratic is $(11 + 2\sqrt{73})/3$, between 9 and 10. The retained integers are 10,11,12.

**Worked derivation:**

$$3(9)^2 - 22(9) - 57 = -12, \quad 3(10)^2 - 22(10) - 57 = 23,$$

and the upward quadratic increases for k at least 7, confirming exactly three retained values.

**Why:** A real-root condition alone does not enforce the reciprocal separation. Positivity justifies dropping absolute values from the product denominator and squaring without changing the inequality. The finite integer range is applied only after all continuous constraints are combined.

**Why not the others:** (A) retains only the largest feasible integer. (B) wrongly excludes the first feasible value k=10. (D) includes k=9 despite its insufficient reciprocal gap.

**Key point:** Convert root-based inequalities into coefficient constraints after checking root signs.

---

**Q44 / A** Two-factor constant-term extraction with cancellation of distinct admissible routes

**Answer:** (A) 0.

**Given:** Two separate binomial expansions contribute signed terms.

**Formula/Principle:** Track exponent and coefficient from both factors; sum every admissible index pair.

**Step 1:** Choosing k inverse-power terms in the first factor gives

$$\binom{8}{k} x^{16-3k}.$$

Choosing j negative inverse-power terms in the second gives

$$(-1)^j \binom{6}{j} x^{6-2j}.$$

**Step 2:** A constant requires

$$22 - 3k - 2j = 0, \quad 0 \le k \le 8, \quad 0 \le j \le 6.$$

**Step 3:** k must be even. Testing its allowed range gives only $(k, j) = (4, 5), (6, 2)$; the other even k give j outside 0 through 6.

**Step 4:** Their contributions are

$$-\binom{8}{4}\binom{6}{5} = -70(6) = -420, \qquad \binom{8}{6}\binom{6}{2} = 28(15) = 420.$$

**Worked derivation:**

$$[x^0](x^2+x^{-1})^8(x-x^{-1})^6 = -420+420 = 0.$$

**Why:** A product can have several independent routes to the same power. The signed second factor makes their sum, rather than their count or magnitudes, decisive. Restricting the index equation to valid binomial ranges prevents spurious terms and reveals exact cancellation.

**Why not the others:** (B) keeps only the positive route. (C) keeps only the negative route. (D) adds absolute magnitudes and discards the alternating sign.

**Key point:** In products of binomial expansions, solve the combined exponent condition and add all signed contributions.

---

**Q45 / B** Odd-index and total GP sums reconstruct a convergent squared-term sum

**Answer:** (B) 72.

**Given:** First term a, ratio r, and both specified convergent geometric sums.

**Formula/Principle:** Odd-indexed terms have ratio r squared; squared terms also form a geometric progression.

**Step 1:**

$$\frac{a}{1-r} = 12, \qquad \frac{a}{1-r^2} = 9.$$

Dividing gives $1+r = 12/9 = 4/3$, so $r = 1/3$.

**Step 2:**

$$a = 12(1-1/3) = 8.$$

**Step 3:**

$$\sum_{n=1}^{\infty} a_n^2 = \frac{a^2}{1-r^2} = \frac{64}{1-1/9} = 72.$$

**Worked derivation:**

$$\sum a_n^2 = \frac{8^2}{1-(1/3)^2} = 72, \qquad |r^2| < 1.$$

**Why:** The supplied odd-index sum constrains the ratio without knowing the first term. Squaring the sequence changes both its first term and ratio; it is not the same operation as squaring the original sum, which would include cross terms. Only geometric-series convergence is used.

**Why not the others:** (A) uses an incorrect squared-term ratio or first term. (C) squares the odd-index sum. (D) squares the total sum instead of summing individual squares.

**Key point:** Form the new progression explicitly when selecting or squaring terms of a GP.

---

## Physics

**Q46 / B** Limiting top-of-circle contact with nonzero gravitational acceleration

**Answer:** (B) Normal force zero, with gravity providing inward acceleration.

**Given:** At the top, both gravity and any compressive track force point towards the centre.

**Formula/Principle:** Apply the radial component of Newton's second law.

**Step 1:**

$$mg + N = mv^2/R.$$

**Step 2:** With $v^2 = gR$,

$$N = mg - mg = 0, \qquad a_r = v^2/R = g.$$

**Worked derivation:**

$$N = m(v^2/R - g) = 0, \quad \mathbf{a}_r \text{ is downward.}$$

**Why:** A vanishing contact force does not imply vanishing net force. This is the limiting speed for maintaining contact at the top, where gravity alone provides the required centripetal acceleration. Centripetal force is the inward net force, not a third force to add to gravity and the normal force.

**Why not the others:** (A) retains a normal force that would make the inward acceleration too large. (C) confuses zero normal force with force-free motion. (D) reverses the top-of-circle directions and doubles a force without balancing the stated speed.

**Key point:** Resolve actual forces radially instead of adding a separate centripetal force.

---

**Q47 / C** Inelastic momentum transfer followed by frictional spring compression

**Answer:** (C) 0.60 m.

**Given:** The collision is perfectly inelastic; subsequent friction and spring forces act during displacement.

**Formula/Principle:** Conserve momentum during impact, then use work-energy after impact.

**Step 1:**

$$(1)(6) = (1+2)v \Rightarrow v = 2 \text{ m/s}.$$

**Step 2:** Post-impact kinetic energy is

$$K = \tfrac{1}{2}(3)(2)^2 = 6 \text{ J}.$$

**Step 3:** At maximum compression x,

$$6 = \tfrac{1}{2}(20)x^2 + 4x \Rightarrow 10x^2 + 4x - 6 = 0.$$

**Step 4:** The roots are x=0.60 m and -1.0 m; only positive compression is physical.

**Worked derivation:**

$$\tfrac{1}{2}(20)(0.60)^2 + 4(0.60) = 3.60 + 2.40 = 6.00 \text{ J}.$$

**Why:** Embedding conserves the brief collision momentum but dissipates mechanical energy. Once impact ends, friction removes additional energy while the spring stores the remainder. Using initial projectile energy directly would count energy already converted to internal forms during impact.

**Why not the others:** (A) is inconsistent with the post-impact energy budget. (B) undercounts the available compression energy. (D) ignores friction after correctly finding the post-impact kinetic energy.

**Key point:** Change conservation models at an inelastic collision before analysing later spring motion.

---

**Q48 / D** Off-centre tunnel projection separates period from maximum speed

**Answer:** (D) Period ratio 1, maximum speed ratio $\sqrt{3}/2$.

**Given:** The planet is uniform; tunnel offset is b=R/2.

**Formula/Principle:** Interior gravity is proportional to radius, and only its component along the constrained tunnel accelerates the bead.

**Step 1:** With s measured along the tunnel from its midpoint,

$$a_s = -(g/R)s.$$

The perpendicular component is balanced by the tunnel constraint.

**Step 2:** Hence

$$\omega = \sqrt{g/R}, \quad T = 2\pi\sqrt{R/g},$$

independent of offset.

**Step 3:** The amplitude is the half-chord length

$$A = \sqrt{R^2 - b^2} = \sqrt{3}R/2.$$

**Step 4:** Since $v_{\max} = \omega A$, its ratio to the diameter-tunnel value $\omega R$ is $\sqrt{3}/2$.

**Worked derivation:**

$$\frac{T_b}{T_0} = 1, \qquad \frac{v_{\max,b}}{v_{\max,0}} = \frac{\sqrt{R^2 - (R/2)^2}}{R} = \frac{\sqrt{3}}{2}.$$

**Why:** The restoring coefficient along every straight chord is the same inside a uniform sphere, even though different chords have different available amplitudes. Equal periods therefore do not imply equal maximum speeds. Surface gravity cannot be treated as a constant force throughout the tunnel.

**Why not the others:** (A) scales period and speed directly with the offset. (B) transfers the amplitude ratio to period and leaves speed unchanged. (C) mistakes the offset for the half-chord length.

**Key point:** Separate the restoring coefficient from amplitude in constrained gravitational oscillations.

---

**Q49 / A** The first stress limit determines the maximum total energy of unequal series wires

**Answer:** (A) $3\sigma^2 AL/(4Y)$.

**Given:** The same tensile force passes through both series wires.

**Formula/Principle:** Find the first limiting force from stress and then sum the elastic energies $F^2 L/(2YA_i)$.

**Step 1:** The thinner wire has the larger stress, so the first limit is

$$F_{\max} = \sigma A.$$

**Step 2:** The extensions at that force are

$$\Delta L_1 = \sigma L/Y, \qquad \Delta L_2 = \sigma L/(2Y).$$

**Step 3:** Total stored energy is

$$U = \tfrac{1}{2} F_{\max}(\Delta L_1 + \Delta L_2) = \frac{3\sigma^2 AL}{4Y}.$$

**Worked derivation:**

$$U = \frac{(\sigma A)^2 L}{2YA} + \frac{(\sigma A)^2 L}{2Y(2A)} = \frac{3\sigma^2 AL}{4Y}.$$

**Why:** Series connection equalises force, not stress or strain. The narrower wire reaches the material limit while the wider wire remains below it. Both still store energy, so evaluating only the limiting wire misses part of the system's energy.

**Why not the others:** (B) includes only the thin wire's energy. (C) omits the one-half factor for reversible linear loading. (D) includes only the wider wire's energy.

**Key point:** Identify which component reaches its local stress limit before summing a system energy.

---

**Q50 / C** A measured junction temperature calibrates a series-to-parallel conduction change

**Answer:** (C) 25/6.

**Given:** Equal L and A; the series temperature drops are 60 and 40 degrees.

**Formula/Principle:** Equal series heat currents determine the conductivity ratio; parallel thermal conductances then add.

**Step 1:**

$$k_1 A(60)/L = k_2 A(40)/L \Rightarrow k_1/k_2 = 2/3.$$

**Step 2:** Series conductance is

$$G_s = \frac{A}{L}\frac{k_1 k_2}{k_1 + k_2},$$

whereas parallel conductance is $G_p = (A/L)(k_1 + k_2)$.

**Step 3:** With the same reservoir difference,

$$\frac{\dot{Q}_p}{\dot{Q}_s} = \frac{(k_1+k_2)^2}{k_1k_2} = \frac{(2+3)^2}{2(3)} = \frac{25}{6}.$$

**Worked derivation:**

$$\dot{Q}_p/\dot{Q}_s = G_p/G_s = 25/6.$$

**Why:** The junction temperature is not an average unless thermal resistances are equal. It first identifies their ratio. Reconfiguration changes both rods' temperature drops from partial drops to the full reservoir difference, so simply adding their original series currents would be incorrect.

**Why not the others:** (A) compares one rod to an incorrect average resistance. (B) inverts the requested ratio. (D) ignores the series combination when normalising the parallel current.

**Key point:** Infer transport coefficients from the original temperature division before changing a thermal network.

---

**Q51 / B** Mean-free-path compression inferred before adiabatic pressure and collision-time ratios

**Answer:** (B) $16$ and $1/(8\sqrt{2})$.

**Given:** Cv=3R, fixed collision cross-section, and mean free path ratio 1/8.

**Formula/Principle:** $\lambda \propto V/N$, $\gamma = (C_v+R)/C_v$, $TV^{\gamma-1}$ = constant, and $\tau \sim \lambda/v_{\text{rms}}$.

**Step 1:** Fixed N and diameter give

$$V_2/V_1 = \lambda_2/\lambda_1 = 1/8.$$

**Step 2:**

$$\gamma = 4/3, \quad T_2/T_1 = (V_1/V_2)^{1/3} = 2.$$

**Step 3:** From the ideal gas law,

$$P_2/P_1 = (T_2/T_1)(V_1/V_2) = 2(8) = 16.$$

**Step 4:** Molecular mass is unchanged, so

$$v_2/v_1 = \sqrt{2}, \quad \tau_2/\tau_1 = \frac{1/8}{\sqrt{2}} = \frac{1}{8\sqrt{2}}.$$

**Worked derivation:**

$$\frac{P_2}{P_1} = 16, \qquad \frac{\tau_2}{\tau_1} = \frac{\lambda_2 v_1}{\lambda_1 v_2} = \frac{1}{8\sqrt{2}}.$$

**Why:** Shorter mean free path measures increased number density, not temperature by itself. The adiabatic condition supplies the temperature change, which then affects both pressure and collision speed. The specified rms-speed time model is used consistently for the ratio; no claim of an exact distribution-averaged collision time is needed.

**Why not the others:** (A) assumes isothermal compression. (C) ignores the density change in pressure and free path. (D) multiplies by the speed ratio instead of dividing.

**Key point:** Combine density, thermodynamic path and speed before estimating a collision time.

---

**Q52 / D** Rapid pendulum shortening changes amplitude through angular momentum rather than energy conservation

**Answer:** (D) $8\theta_0$.

**Given:** The pull occurs at the lowest point and changes length instantaneously on the oscillation time scale.

**Formula/Principle:** Use angular momentum for the pull, then small-angle energy or SHM for the new oscillation.

**Step 1:** Initially

$$v_0 = \theta_0\sqrt{gL}.$$

**Step 2:** Angular momentum conservation gives

$$mLv_0 = m(L/4)v_1 \Rightarrow v_1 = 4v_0.$$

**Step 3:** At new length $L' = L/4$, the lowest-point speed is $v_1 = \theta_1\sqrt{gL'}$. Thus

$$\theta_1 = \frac{4\theta_0\sqrt{gL}}{\sqrt{gL/4}} = 8\theta_0.$$

**Worked derivation:**

$$mL^2\dot{\theta}_0 = m(L/4)^2\dot{\theta}_1, \quad \omega_1 = 2\omega_0, \quad \theta_1/\theta_0 = 8.$$

**Why:** The radial pulling force has no torque about the pivot but does mechanical work, so angular momentum rather than mechanical energy is conserved during shortening. After the pull the new pendulum converts its increased kinetic energy into gravitational potential energy. The small-angle assumption must hold even for the amplified angle.

**Why not the others:** (A) uses only the square-root frequency scaling. (B) uses the speed ratio as the angular-amplitude ratio and ignores the changed length. (C) assumes energy conservation during an externally powered pull.

**Key point:** A torque-free radial intervention can conserve angular momentum while changing oscillation energy.

---

**Q53 / C** A charged isolated middle node invalidates ordinary neutral-series voltage division

**Answer:** (C) M is at V and total energy is $CV^2/2$.

**Given:** A=0, B=V and the middle conductor has prescribed nonzero free charge.

**Formula/Principle:** Add the charges on both capacitor plates belonging to M, then calculate energy from each actual voltage.

**Step 1:** If M has potential Vm, its two plate charges sum to

$$Q_M = CV_m + 2C(V_m - V) = 3CV_m - 2CV.$$

**Step 2:** Setting $Q_M = CV$ gives

$$V_m = V.$$

**Step 3:** The C capacitor has voltage V; the 2C capacitor has zero voltage. Therefore

$$U = \tfrac{1}{2}CV^2 + \tfrac{1}{2}(2C)(0)^2 = \tfrac{1}{2}CV^2.$$

**Worked derivation:**

$$Q_{M,\text{left}} = CV, \quad Q_{M,\text{right}} = 0, \quad U = CV^2/2.$$

**Why:** Equal-magnitude series charges require an initially neutral isolated junction. Here its specified charge changes the voltage division completely. The source fixes the end-node potentials while junction-charge conservation fixes the middle potential; both constraints must be used.

**Why not the others:** (A) applies the neutral-junction series formula. (B) reverses the charge contribution in the middle-node balance. (D) assigns voltage V to both capacitors even though M and B are equipotential.

**Key point:** Write a junction's net-charge equation before assuming equal series-capacitor charges.

---

**Q54 / A** Instant field reversal gives zero torque at an unstable energy maximum

**Answer:** (A) $+\mu B$, zero torque and unstable.

**Given:** The loop has not rotated when the field reverses, so its angle to the new field is pi.

**Formula/Principle:** $U = -\mu B \cos\theta$, $|\tau| = \mu B|\sin\theta|$; local energy curvature determines stability.

**Step 1:**

$$U(\pi) = +\mu B, \qquad |\tau(\pi)| = 0.$$

**Step 2:** A small displacement epsilon gives

$$U(\pi + \epsilon) = \mu B \cos\epsilon < \mu B$$

for small nonzero epsilon, so the configuration is an energy maximum and unstable.

**Worked derivation:**

$$\left.\frac{d^2U}{d\theta^2}\right|_{\pi} = \mu B \cos\pi = -\mu B < 0.$$

**Why:** An energy maximum can have zero instantaneous torque at its exact orientation. Field reversal changes the potential-energy landscape but, by assumption, does not instantly rotate or accelerate the loop through a finite angle. A perturbation then produces a torque away from the anti-aligned state.

**Why not the others:** (B) retains the old aligned energy and stability after reversal. (C) assigns maximum torque to the angle where sine vanishes. (D) confuses the perpendicular orientation with the anti-aligned one.

**Key point:** Zero torque can describe stable or unstable equilibrium; inspect the energy near the orientation.

---

**Q55 / D** Ring-field curvature yields a strict axial stability threshold

**Answer:** (D) $a > R/\sqrt{2}$.

**Given:** Both charges and the test charge are positive; motion is constrained to the common axis.

**Formula/Principle:** Symmetry gives zero field at the midpoint; the derivative of the total field sets linear stability.

**Step 1:**

$$E(x) = kQ\left[\frac{x-a}{(R^2+(x-a)^2)^{3/2}} + \frac{x+a}{(R^2+(x+a)^2)^{3/2}}\right].$$

**Step 2:** Differentiating $s(R^2+s^2)^{-3/2}$ gives $(R^2-2s^2)/(R^2+s^2)^{5/2}$. Hence

$$E'(0) = \frac{2kQ(R^2-2a^2)}{(R^2+a^2)^{5/2}}.$$

**Step 3:** For q>0, restoring force requires $qE'(0) < 0$, so

$$R^2 - 2a^2 < 0 \Rightarrow a > R/\sqrt{2}.$$

**Worked derivation:**

$$F(x) \simeq qE'(0)x, \qquad F(x)x < 0 \iff a > R/\sqrt{2}.$$

**Why:** Midpoint cancellation alone does not establish stability. The slope decides whether a displaced positive charge is pushed back or further away. The equality case has zero linear term and therefore is not a nonzero linear restoring force; the question does not claim three-dimensional electrostatic trapping.

**Why not the others:** (A) has a positive field slope and is axially unstable. (B) superposes rings at one plane, also giving positive axial slope. (C) marks the vanishing-linear-term boundary rather than the required strict region.

**Key point:** Differentiate a symmetric field to distinguish equilibrium from linear restoring behaviour.

---

**Q56 / B** Loaded-voltmeter calibration followed by external-power optimisation

**Answer:** (B) 12.5 W.

**Given:** The meter loads the cell in both calibration measurements and is absent in the final optimisation.

**Formula/Principle:** Combine the meter and load in parallel, infer E and r from terminal voltages, then maximise external power.

**Step 1:** The effective loads are

$$R_{e1} = 6 \parallel 6 = 3 \text{ ohm}, \quad R_{e2} = 3 \parallel 6 = 2 \text{ ohm}.$$

**Step 2:** Currents are $I_1 = 6/3 = 2$ A and $I_2 = 5/2 = 2.5$ A. Thus

$$E = 6 + 2r = 5 + 2.5r \Rightarrow r = 2 \text{ ohm}, \quad E = 10 \text{ V}.$$

**Step 3:** With the meter removed,

$$P(R) = \frac{E^2 R}{(R+r)^2}.$$

Its derivative has the sign of r-R, so the unique maximum is at R=r.

**Step 4:**

$$P_{\max} = \frac{E^2}{4r} = \frac{100}{8} = 12.5 \text{ W}.$$

**Worked derivation:**

$$P'(R) = E^2 \frac{r-R}{(R+r)^3}, \quad R_{\max} = 2 \text{ ohm}, \quad P_{\max} = 12.5 \text{ W}.$$

**Why:** Terminal voltage is not emf when current flows, and the finite-resistance meter contributes additional current. Ignoring that loading corrupts both inferred cell parameters. The final circuit is a different load arrangement, so its power is optimised after removing the instrument.

**Why not the others:** (A) underestimates the maximum by a factor two. (C) uses an incorrect denominator in the matched-load power. (D) counts internal and external dissipation as externally delivered power at the optimum.

**Key point:** Correct instrument loading before using inferred source parameters in a new circuit.

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**Q57 / D** Two opposite-sign magnifications determine an unknown focal length

**Answer:** (D) 8 cm.

**Given:** Signed magnifications are m1=-2 and m2=+4; real objects lie to the left.

**Formula/Principle:** Use $1/f = 1/v - 1/u$ and $m = v/u$, so $u = f(1/m - 1)$.

**Step 1:**

$$u_1 = f(-1/2 - 1) = -3f/2.$$

**Step 2:**

$$u_2 = f(1/4 - 1) = -3f/4.$$

**Step 3:** Moving towards the lens increases u by 6 cm:

$$u_2 - u_1 = 3f/4 = 6 \Rightarrow f = 8 \text{ cm}.$$

**Worked derivation:**

$$u_1 = -12 \text{ cm}, \; v_1 = 24 \text{ cm}; \quad u_2 = -6 \text{ cm}, \; v_2 = -24 \text{ cm}.$$

**Why:** The image switches from real inverted to virtual upright because the object crosses the focal point. Using unsigned magnifications loses this distinction and can produce an invalid focal length. Both reconstructed positions satisfy the same lens equation and the observed displacement.

**Why not the others:** (A) halves the result by an incorrect displacement relation. (B) equates focal length with object travel. (C) reports the initial object distance rather than focal length.

**Key point:** Use signed magnification when a displacement changes image type.

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**Q58 / A** A common photon-energy change leaves a two-metal stopping-potential difference invariant

**Answer:** (A) B has a 1.5 eV larger work function and the voltage difference is invariant.

**Given:** Both metals receive photons of the same energy before and after an equal frequency change.

**Formula/Principle:** Einstein's equation is $K_{\max} = hf - \phi$, with $eV_s = K_{\max}$.

**Step 1:** Subtract the two equations:

$$\phi_B - \phi_A = K_A - K_B = 2.0 - 0.5 = 1.5 \text{ eV}.$$

**Step 2:** A common frequency increase adds the same $h\Delta f$ to both kinetic energies, leaving their difference 1.5 eV.

**Worked derivation:**

$$V'_{s,A} - V'_{s,B} = \frac{(K_A + h\Delta f) - (K_B + h\Delta f)}{e} = 1.5 \text{ V}.$$

**Why:** The metal producing the slower maximum-energy electrons spends more photon energy overcoming its work function. A shared photon-energy increase cancels on subtraction. Stopping potential measures maximum kinetic energy per charge, so its difference is invariant even though each individual stopping potential rises.

**Why not the others:** (B) reverses the work-function ordering. (C) ignores unequal electron energies under identical illumination. (D) adds the kinetic energies and incorrectly scales a difference that cancels the common increment.

**Key point:** Subtract photoelectric equations to isolate material differences from common illumination.

---

**Q59 / C** Head-on Coulomb turning distance interpreted as a size bound

**Answer:** (C) 20 fm and an upper scale bound.

**Given:** Alpha charge is +2e, target charge +50e, and the approach is head-on with negligible target recoil.

**Formula/Principle:** At closest approach all initial kinetic energy is Coulomb potential energy.

**Step 1:**

$$K = \frac{k(2e)(50e)}{r_{\min}}.$$

**Step 2:**

$$r_{\min} = \frac{100(1.44)}{7.2} = 20 \text{ fm}.$$

**Step 3:** Coulomb behaviour without resolved nuclear-size effects down to this separation places the nuclear size below this probing scale within the model; it does not prove equality.

**Worked derivation:**

$$7.2 \text{ MeV} = \frac{144 \text{ MeV fm}}{20 \text{ fm}}.$$

**Why:** A head-on trajectory reaches its turning point by energy conservation. The closest approach is a probe limitation set by beam energy and charges. A nucleus can be much smaller while producing the same external Coulomb scattering, so identifying the turning point with the nuclear surface is unjustified.

**Why not the others:** (A) omits the alpha particle's charge factor two and also asserts equality. (B) calculates the turning distance correctly but overinterprets it as a surface measurement. (D) doubles the turning distance and reverses the size-bound interpretation.

**Key point:** Distinguish a scattering experiment's closest probe distance from a measured object radius.

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**Q60 / B** Stored magnetic energy partitioned between series dissipative paths

**Answer:** (B) 6.75 J.

**Given:** Only the inductor initially stores energy; both resistors carry the same transient current.

**Formula/Principle:** Total heat equals lost magnetic energy, and instantaneous resistor powers divide in the ratio of resistances.

**Step 1:**

$$U_i = \tfrac{1}{2}LI_0^2 = \tfrac{1}{2}(2)(3)^2 = 9 \text{ J}.$$

**Step 2:** At every time,

$$P_6/P_2 = 6i^2/(2i^2) = 3,$$

so the same ratio holds for integrated heats.

**Step 3:**

$$Q_6 = \frac{6}{2+6}U_i = \frac{3}{4}(9) = 6.75 \text{ J}.$$

**Worked derivation:**

$$Q_2 + Q_6 = 9 \text{ J}, \quad Q_2 = 2.25 \text{ J}, \quad Q_6 = 6.75 \text{ J}.$$

**Why:** Disconnecting the source does not erase the magnetic energy instantly. It drives a decaying current through the retained closed path, dissipating energy in both resistors. The time dependence cancels in the energy fraction because their series current is identical at every instant.

**Why not the others:** (A) assigns all stored energy to one resistor. (C) gives the heat in the 2 ohm resistor. (D) omits the one-half factor in magnetic energy and also ignores the sharing.

**Key point:** Use common-current power ratios to partition a transient energy loss without solving the full current history.

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